

INTRODUCTION

Oral health is a fundamental component of overall health and well-being, directly impacting an individual's quality of life [1]. As the healthcare industry modernizes, the adoption of specialized health information technology has become essential for clinical efficiency and improved patient care [2]. Dental Information Management Systems have emerged to help transition clinics from traditional to streamlined digital environments [3]. These systems facilitate the secure management of Electronic Medical Records (EMR), patient history, inventory, and scheduling, thereby reducing administrative burdens and allowing dental practitioners to focus primarily on patient treatment [4].

Despite growing technology and digitalization, many startup and expanding dental clinics still rely on manual methods for their daily operations [2]. This often results in disorganized patient records, scheduling conflicts, and inefficient clinic management [4]. Furthermore, patients frequently experience anxiety or hesitation due to a lack of accessible, personalized dental health information [5]. Integrating Artificial Intelligence (AI) for educational patient engagement and recommended scheduling addresses this gap, providing patients with a clearer understanding of their oral health and fostering higher trust and satisfaction [6].

Addressing these operational and patient engagement challenges is a highly necessary and impactful step for growing dental practices. Developing an integrated platform that automates clinic workflows, such as centralizing electronic medical records (EMR), modernizing appointment scheduling, and utilizing artificial intelligence for patient education and recommended scheduling, can significantly resolve existing operational problems. Various studies provided evidence that the implementation of specialized information systems across different medical and dental clinics has greatly improved overall clinical efficiency [3], reduced administrative errors [4], and elevated the quality of patient care [7]. Therefore, applying these technological innovations is important in bridging the gap between traditional manual operations and modernized dental healthcare delivery.

1.1 PROJECT CONTEXT

The United Nations' Sustainable Development Goal 3 emphasized the importance of ensuring healthy lives and promoting well-being for all at all ages [8]. A key factor in achieving this goal is the accessibility, efficiency, and quality of healthcare delivery, including oral health. To fulfill

this objective globally, the healthcare industry is actively transitioning from traditional practices to digital health information systems to improve patient outcomes and clinic operations [7]. However, in developing countries like the Philippines, dental practice modernization remains in transition. While top-tier hospitals have fully embraced digitalization, many local and expanding dental clinics still encounter barriers to digital transformation, leading to operational inefficiencies that directly affect patient care and clinic management [9]. Locally, expanding a dental clinic into a multi-branch enterprise introduces significant operational challenges. When clinics rely heavily on manual procedures and fragmented communication platforms, they face issues such as isolated data silos and insecure transfer of sensitive medical records. Without unified digital infrastructure, both the administrative efficiency of the clinic and the overall engagement and satisfaction of the patients are compromised [10].

Dentime Dental Clinic, established in 2024, is a rapidly growing dental practice located at 2/F JN Terraces Bldg., 881 J. P. Rizal Street, Brgy. Concepcion Uno, Marikina City, operating with a floor area of 30 square meters. Dr. Michael Edwin Tamayo, DMD, owns the clinic. Currently, the clinic has one in-house dentist, one secretary, and a clinic manager, and it exclusively offers dental services. Serving about 6 to 10 patients daily through scheduled appointments and walk-ins, the clinic has seen a steady patient flow. This inspired management to expand operations to accommodate and treat more patients from other communities by constructing a second branch at 2/F Arquillos Commercial Bldg., Phase 1 Block 12 Lot 3 Eastwood Residences, San Isidro, Rodriguez, Rizal. To support the expansion, management plans to hire additional dentists as needed, depending on the initial patient volume at the new location. While there are no concrete plans for additional branches yet, the management estimates that future clinics would also require around 30 to 50 square meters of space to support their operations effectively.

Currently, Dentime Dental Clinic relies on manual, fragmented methods to manage daily operations. The clinic accommodates walk-ins and schedules patient appointments primarily through its Facebook Page and Messenger. The clinic then tracks these appointments using Google Calendar and Excel spreadsheets. The clinic stores patient histories and dental records manually in paper files and folders. Furthermore, the clinic does not formally record dental materials used during treatments, relying only on visual checks.

As the clinic serves more patients, its current appointment setup creates significant challenges. While the clinic manages to avoid overlapping schedules, relying on different platforms

like Facebook Messenger for appointments and follow-up reminders, alongside Google Calendar and Excel for tracking, requires significant manual effort from the limited staff. This fragmented communication and tracking setup compromises optimal clinical efficiency.

Furthermore, the ongoing construction of the second branch highlights a critical limitation in records management. With dental records currently stored in physical folders, transferring patient histories and treatment records between the main clinic and the new branch will have to be done manually. Without a centralized digital database, retrieving and sharing these physical records across locations will cause operational delays, risk data misplacement, and lead to fragmented medical histories for patients visiting multiple branches.

Another significant operational challenge is manually monitoring clinic inventory. Currently, the clinic relies solely on visual checks, with no formal record to track the use of fast-moving and perishable dental supplies such as cotton, anesthesia, brackets and wires for braces, needles, syringes, gloves, and masks. Without system-assisted inventory tracking, staff struggle to accurately monitor stock levels after treatments, increasing the risk of supply shortages during critical procedures.

To address these issues, the researchers propose developing NgitiFy, a centralized Dental Information Management System. NgitiFy will replace the clinic's fragmented scheduling process with a dedicated Appointment and Schedule Management Module, allowing for seamless booking and automated notifications. Furthermore, the system will introduce a centralized Electronic Medical Record (EMR) module capable of secure, branch-level data isolation. This ensures that patient files can be safely accessed and transferred between the first and second branches without relying on insecure messaging platforms [3]. Additionally, the system includes a comprehensive dashboard and report generator to track daily patient volume, providing management with essential data to support and justify decisions about future branch expansion.

Beyond administrative efficiency, NgitiFy provides Dentime Dental Clinic with a strong competitive advantage through its AI-Driven Patient Engagement Module. To promote patient awareness and proactive oral care, the system features AI Patient Engagement that provides personalized dental health advice, such as dietary recommendations for teeth maintenance, as well as recommended visit windows for follow-up schedules. Moreover, the AI-Assisted Radiograph Review feature within the Patient EMR allows dentists to review radiograph quality, record clinical

findings and annotations, and prepare dentist-approved summaries for patients. Lastly, the AI Patient Engagement Module supports continuous oral health management by providing access to NgitiBot, Dental Health Education, Oral Health Management, and the Recommended Visit Window, helping patients maintain awareness of their daily oral health. Ultimately, NgitiFy will empower Dentime Dental Clinic to transition from manual, single-branch operations to a highly efficient, multi-branch digital ecosystem, ensuring high-quality service delivery as they continue to expand.

1.2 PURPOSE AND DESCRIPTION

The project, NgitiFy: A Dental Information Management System with AI-Driven Patient Engagement for Dentime Dental Clinic, aims to develop a centralized digital platform that improves patient record management, clinic operations, and patient engagement. This project addresses the clinic's current reliance on manual, non-integrated systems such as walk-in scheduling, Facebook inquiries, and Google Calendar tracking. These methods often create inefficient workflows, fragmented patient histories, and challenges in monitoring appointments and managing clinic inventory. By implementing NgitiFy, the clinic can centralize patient data, automate administrative tasks, and improve overall efficiency, especially as Dentime Dental Clinic expands to multiple branches.

NgitiFy introduces several relevant features, including an Electronic Medical Record, Interactive Digital Odontogram, AI-Assisted Radiograph Review, and AI Patient Engagement Module. The AI Patient Engagement Module consists of NgitiBot, Dental Health Education, Oral Health Management, and the Recommended Visit Window. NgitiBot provides supported conversational oral health guidance and personalized dietary recommendations. Dental Health Education provides approved educational materials, while Oral Health Management allows patients to record and monitor relevant oral health information. The Recommended Visit Window provides visit guidance based on available patient records and oral health inputs. The system also includes appointment scheduling and supply and stock monitoring to improve clinic operations.

The system is comprehensively designed to improve service delivery and administrative operations. Specifically, the development and implementation of NgitiFy will be highly beneficial to the following:

Dental Patients. It provides convenient appointment booking, access to treatment history, dental health educational materials, oral health monitoring tools, and recommended visit guidance, improving overall access to clinic services and oral health information.

Dentists. The system improves clinical workflow by providing secure access to patient records, an interactive digital odontogram, AI-assisted radiograph review with dentist-approved summaries, and accurate material-usage tracking.

Clinic Secretary. Helps efficiently encode patient information, manage appointments and schedules, monitor inventory alerts, and handle patient concerns, significantly reducing manual workload and facilitating secure data transfers.

Branch Manager. Enables oversight of branch-level operations, including appointment scheduling, queue management, patient registration, and inventory monitoring within the assigned clinic branch.

Clinic Owner. Enables centralized monitoring of daily patient volumes, clinic operations, patient reports, and overall clinic performance across branches for better management and decision-making.

System Administrator. Maintains full control over the NgitiFy system, including user account management for all roles, branch setup, system configuration, database integrity, audit trail monitoring, and role-based access permissions.

Researchers. Enhances knowledge in developing web and mobile healthcare systems, providing valuable learning experiences in artificial intelligence integration, database management, UI/UX design, and software development.

Future Researchers. Serves as a solid foundation and useful reference for future studies on innovations in dental and healthcare management systems.

1.3 OBJECTIVES

General Objective

The primary goal is to develop and implement NgitiFy: A Dental Information Management System with AI-Driven Patient Engagement for Dentime Dental Clinic through a web and mobile application.

Specific Objectives

Specifically, these are the objectives of the capstone project:

1. To develop a centralized web and mobile Electronic Medical Record (EMR) system that ensures the secure synchronization and transfer of patient profiles, medical and dental histories, treatment history, digital odontograms, and radiograph images across multiple branches, featuring role-based access control restricted to the owner, authorized dentists, and clinic staff.
2. To develop an Appointment and Schedule Management module integrated with a Smile Hub for patients that centralizes reservations from various channels, automates notification systems, and features AI-Driven Patient Engagement for personalized NgitiBot, dental health education, overall oral health management, and recommended visit windows.
3. To create an Interactive Digital Odontogram integrated with an AI-Assisted Radiograph Review feature that enables dentists to examine radiograph quality, record and annotate clinical findings, link relevant tooth and treatment information, and prepare dentist-approved summaries for patient reference.
4. To create a Supply and Stock Monitoring module that tracks the inventory levels of perishable dental items across branches and provides automated low-stock alerts. This module allows dentists to log material usage post-treatment for real-time manual deduction. It includes a report-generator dashboard that tracks daily patient volume to help management make branch-expansion decisions.
5. To evaluate the system based on the following criteria:
 - a. Functionality
 - b. Reliability
 - c. Performance Efficiency

- d. Compliance
- e. Usability

1.4 SCOPE AND DELIMITATION

SCOPE

The NgitiFy will function as a centralized web and mobile application, enhancing management operations at Dentime Dental Clinic and its authorized locations. The system will streamline patient record management, appointment scheduling, clinical support functions, and resource tracking. The system will process Electronic Medical Records (EMR), including patient demographic information, medical and dental records, treatment documentation, and digital odontograms. The system will securely distribute records across branches with access controls that permit only admins, owners, branch managers, dentists, and secretaries to access the information.

The system enforces a role-based access control hierarchy: System Administrator, Clinic Owner, Branch Manager, Dentist, Clinic Secretary, and Patient. The System Administrator holds full system-wide access and is responsible for all technical configurations, user account management, and system integrity. The Clinic Owner has cross-branch operational visibility, while the Branch Manager oversees day-to-day operations only within their assigned branch.

NgitiFy will provide clinical support through its Interactive Digital Odontogram system, which enables dentists to document their findings while tracking complete tooth history. The system includes an AI-Assisted Radiograph Review feature that presents objective image-quality information and allows dentists to record clinical findings and prepare review summaries. This feature serves only as decision support and does not independently interpret radiographs or provide a medical diagnosis. The system includes an AI Patient Engagement Module composed of NgitiBot, Dental Health Education, Oral Health Management, and the Recommended Visit Window. NgitiBot serves as the AI Patient Care Companion by providing conversational oral health guidance and personalized dietary recommendations within its supported scope. Dental Health Education provides approved educational materials, while Oral Health Management allows patients to record and monitor relevant oral health information. The Recommended Visit Window provides visit guidance based on available patient records and oral health inputs.

The Appointment and Schedule Management Module will replace traditional calendar systems by centralizing appointment requests, patient schedules, dentist availability, and staff schedules. The system will provide a Smile Hub for patients together with the AI Patient Engagement Module. Within

this module, NgitiBot serves as the AI Patient Care Companion, providing real-time responses to supported patient inquiries. NgitiFy will include a Supply and Stock Monitoring module that monitors perishable dental supplies across different locations. This feature allows dentists to manually log material usage, enabling the system to automatically track deductions and send low-inventory notifications, while also producing daily patient reports that help assess the need for business growth. The system will also be evaluated using an evaluation instrument adapted from the ISO 9126 software quality model and Jakob Nielsen's usability attributes. The assessment covers functionality, reliability, performance efficiency, compliance, and usability. NgitiFy's complete dental information management system provides an efficient platform that supports clinical decision-making while improving the patient experience.

SYSTEM FLOWCHART

In this study, system flowcharts are presented in a swimlane format with two columns: the left column represents user actions, and the right column represents the corresponding processes executed by the system. This format clearly distinguishes each party's responsibilities at every step of a given workflow. Every flowchart begins with the user as the initiating actor, consistent with NgitiFy's event-driven nature, where a user action sets all system processes in motion. A total of forty-eight (48) system flowcharts were constructed to document the complete operational flow of each module and feature of NgitiFy: A Dental Information Management System with AI-Driven Patient Engagement for Dentime Dental Clinic, covering processes ranging from user authentication and patient registration to AI-assisted dental tools, inventory management, and system administration functions.

NGITIFY MOBILE ACTIVITY DIAGRAM

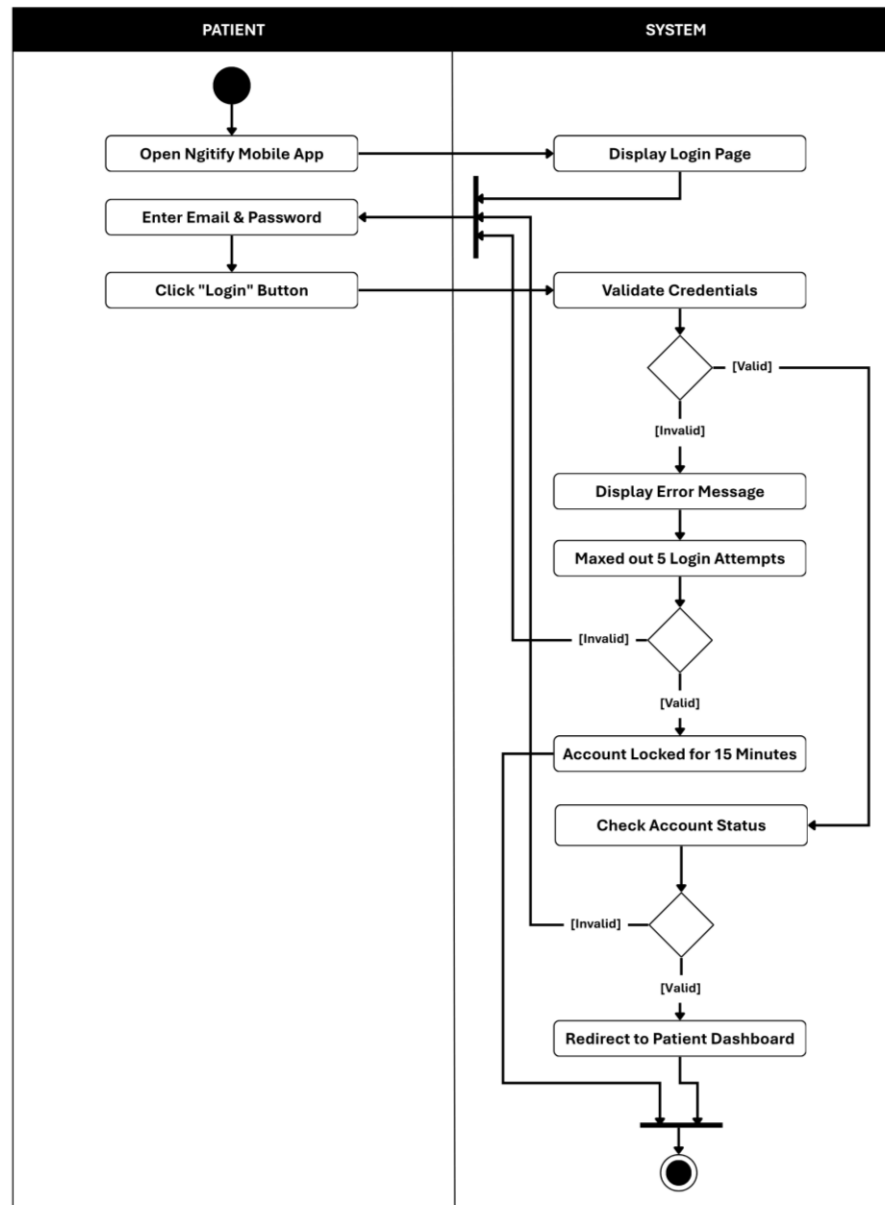


Figure 1. Login Activity Diagram: Mobile

Figure 12 illustrates the mobile login process. The patient opens the NgitiFy mobile application, enters an email address and password, and clicks the login button. The system validates the credentials, checks login attempts and account status, displays errors when needed, locks the account after repeated failed attempts, and redirects the patient to the dashboard when login is successful.

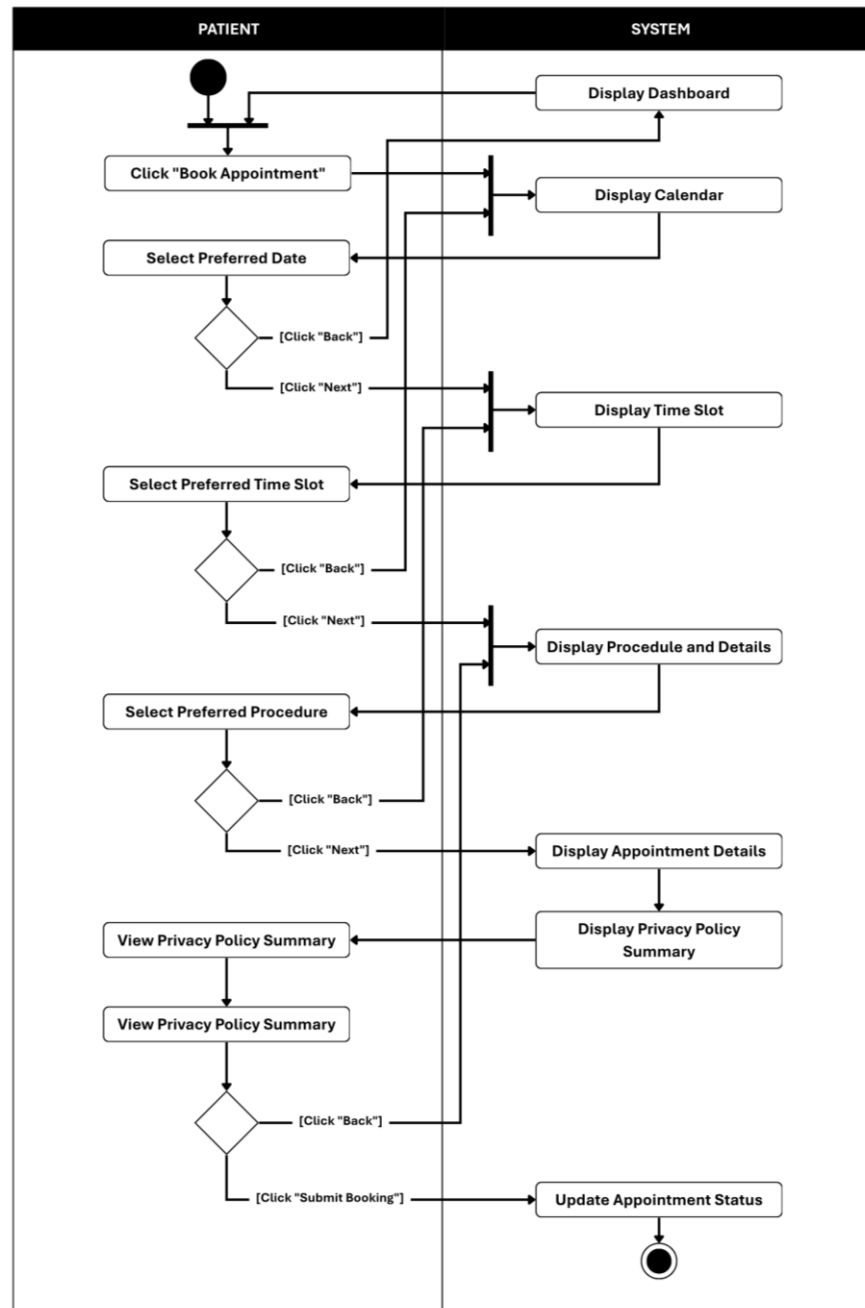


Figure 2. Appointment Scheduling Activity Diagram: Mobile

Figure 2 illustrates how a patient schedules an appointment through the mobile application. The patient selects a preferred date, time slot, and procedure, reviews the appointment and privacy policy summary, and submits the booking request. The system displays the calendar, validates each selection, shows appointment details, and updates the appointment status after submission.

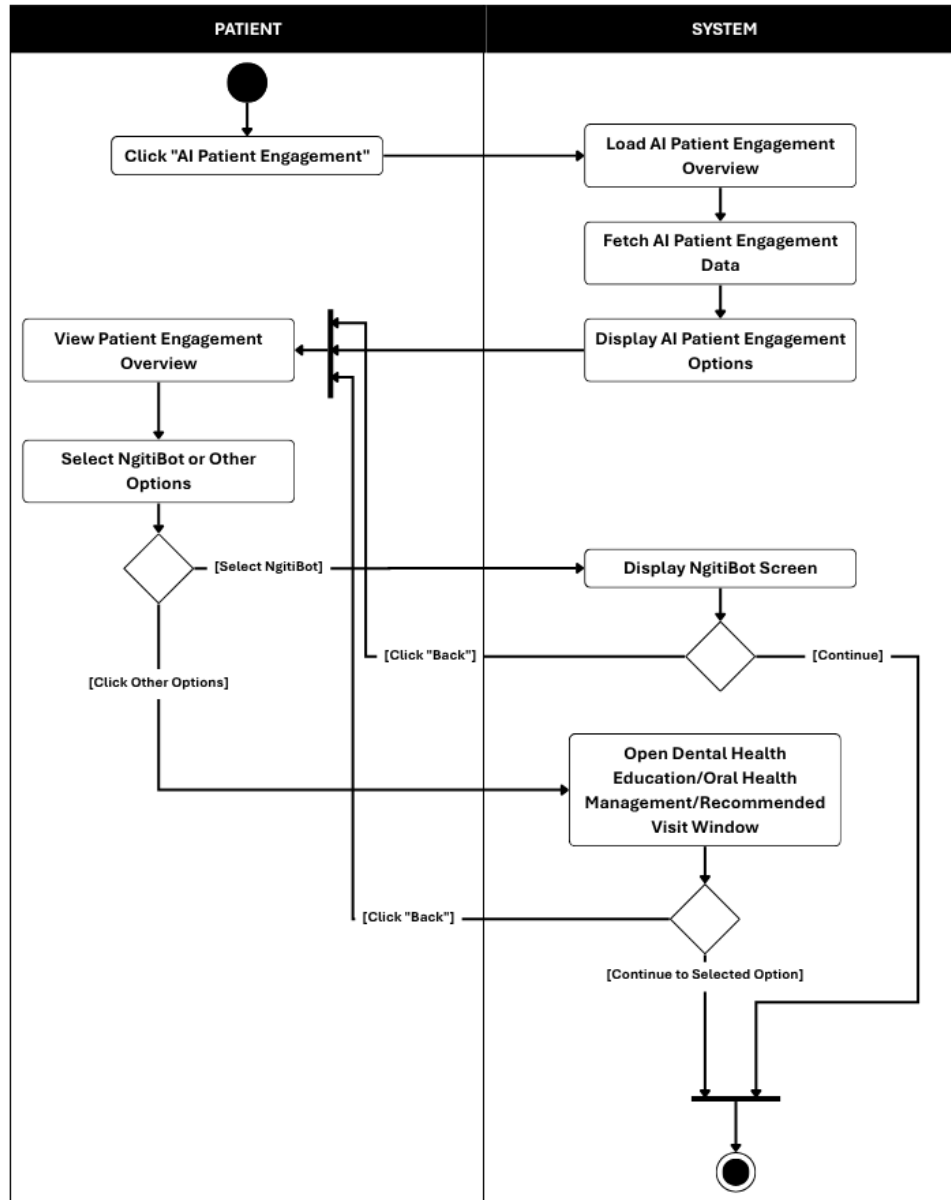


Figure 3. AI Patient Engagement Activity Diagram: Mobile

Figure 3 illustrates how the patient accesses AI Patient Engagement in the mobile application. The system loads the engagement overview, retrieves patient-related data, displays available options, and routes the patient to NgitiBot, dental health education, oral health management, or recommended visit features depending on the selected option.

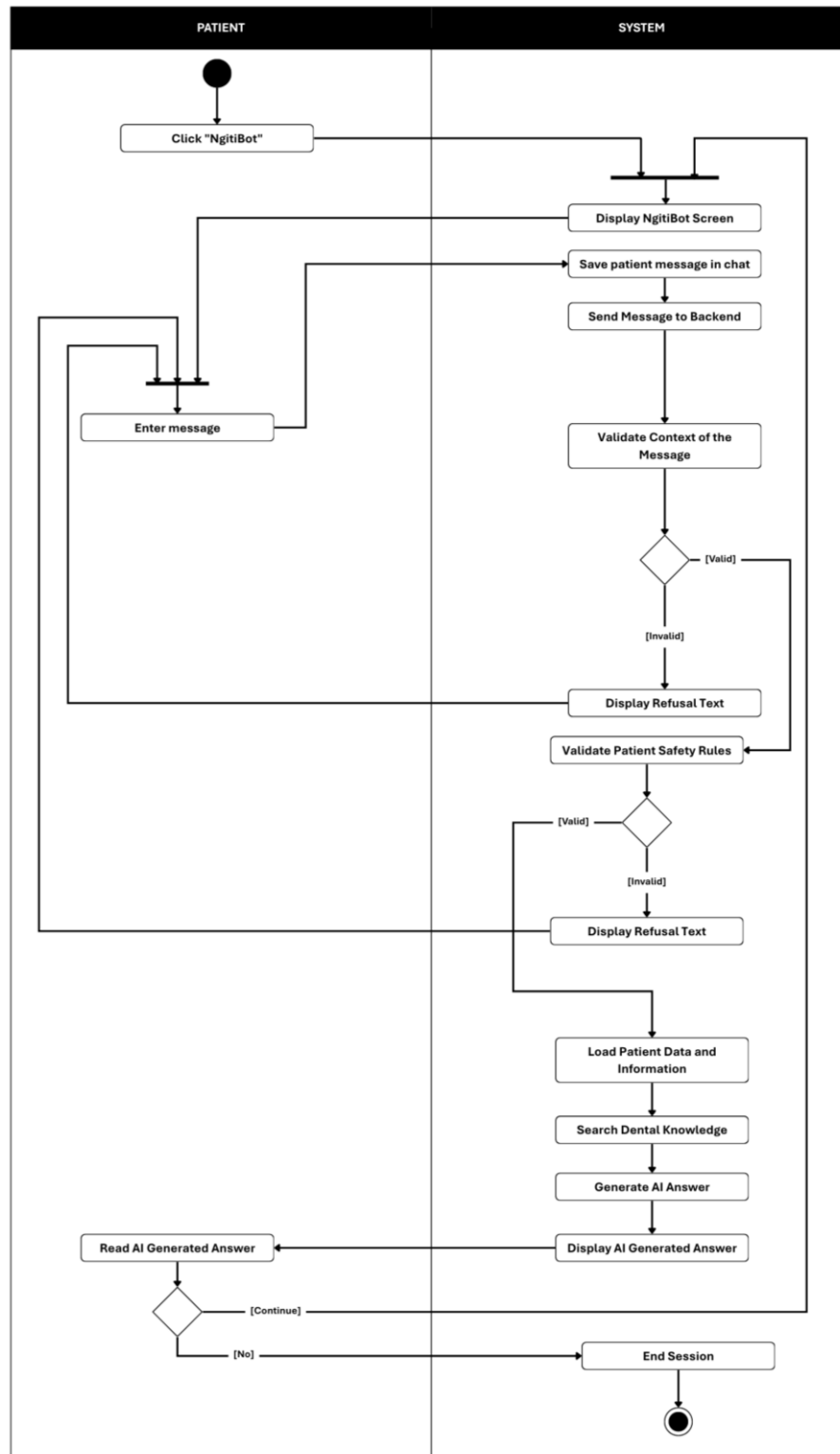


Figure 4. NgitiBot Activity Diagram: Mobile

Figure 4 illustrates the NgitiBot conversation process in the mobile application. The patient enters a message, and the system saves the message, sends it to the backend, validates the content and patient safety rules, retrieves relevant patient and dental knowledge data, generates an AI response, and displays either the generated answer or a refusal message when the inquiry is outside the allowed scope.

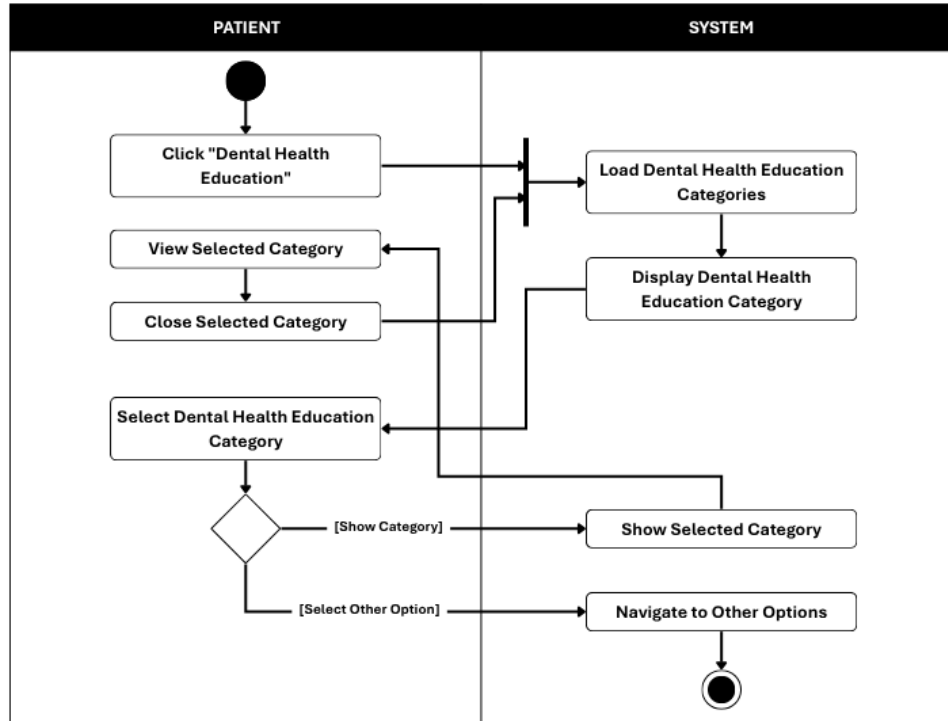


Figure 5. Dental Health Education Activity Diagram: Mobile

Figure 16 illustrates how the patient uses the dental health education feature. The patient opens Dental Health Education, views the available categories, selects a topic, and the system displays the selected educational content or redirects the patient to another available option.

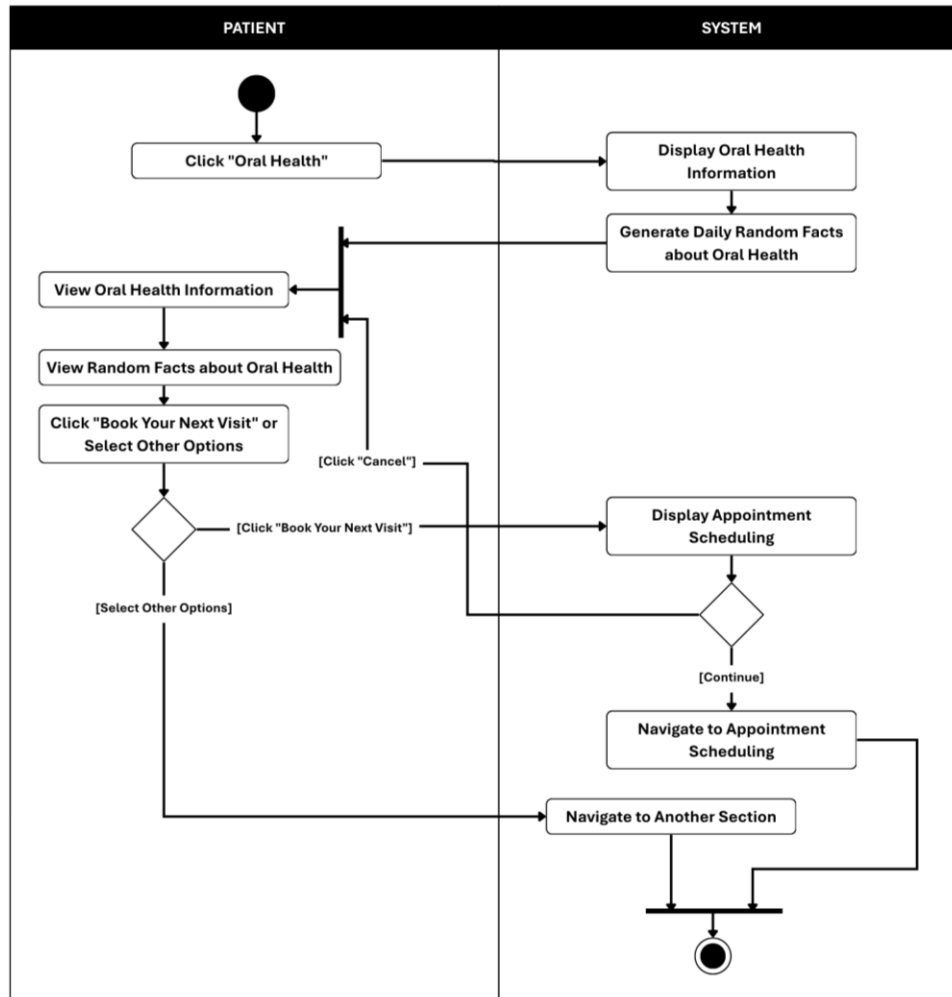
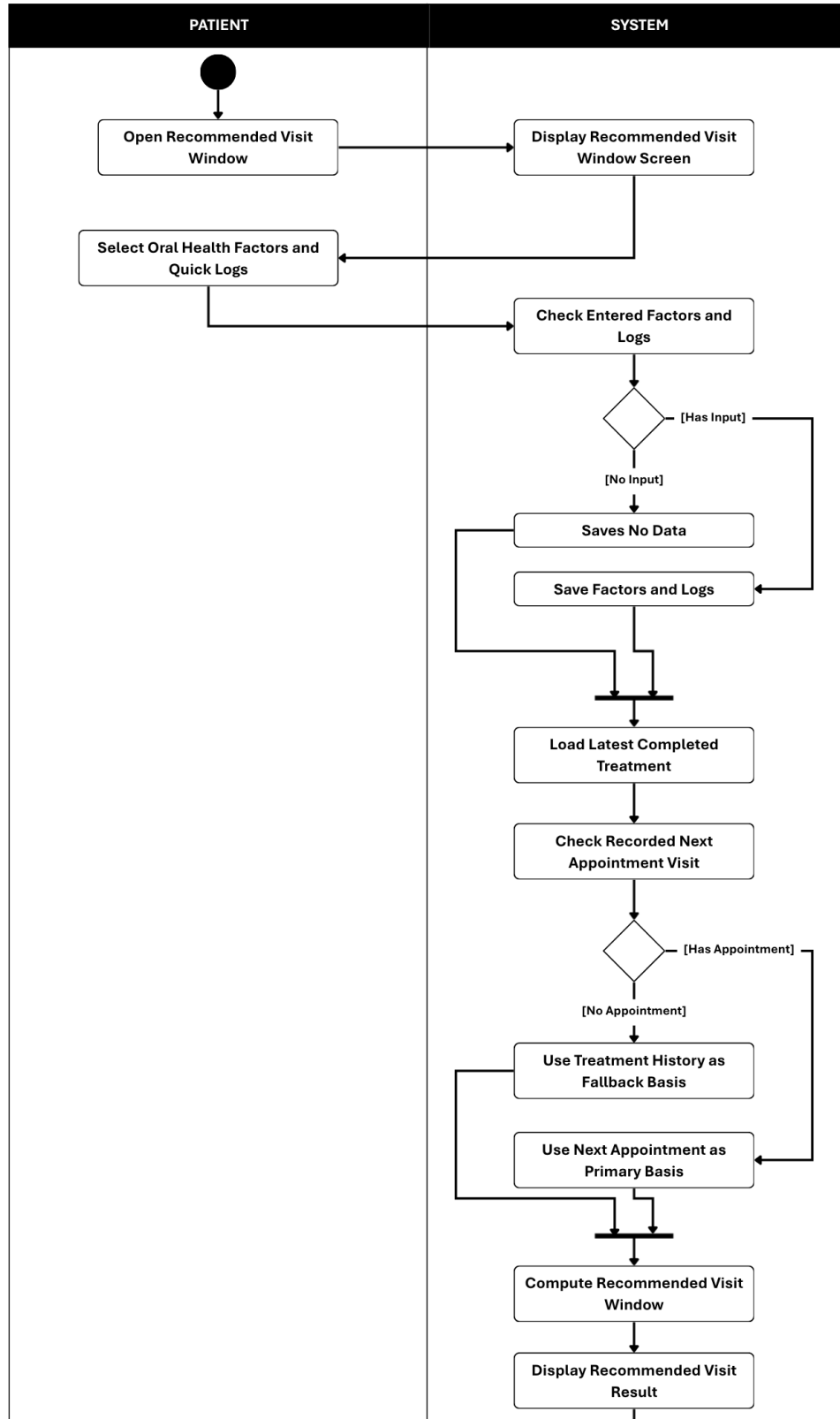


Figure 6. Oral Health Management Activity Diagram: Mobile

Figure 17 illustrates the oral health management process. The system displays oral health information and daily random facts, and the patient can view oral health content, book the next visit, or navigate to another section of the application.



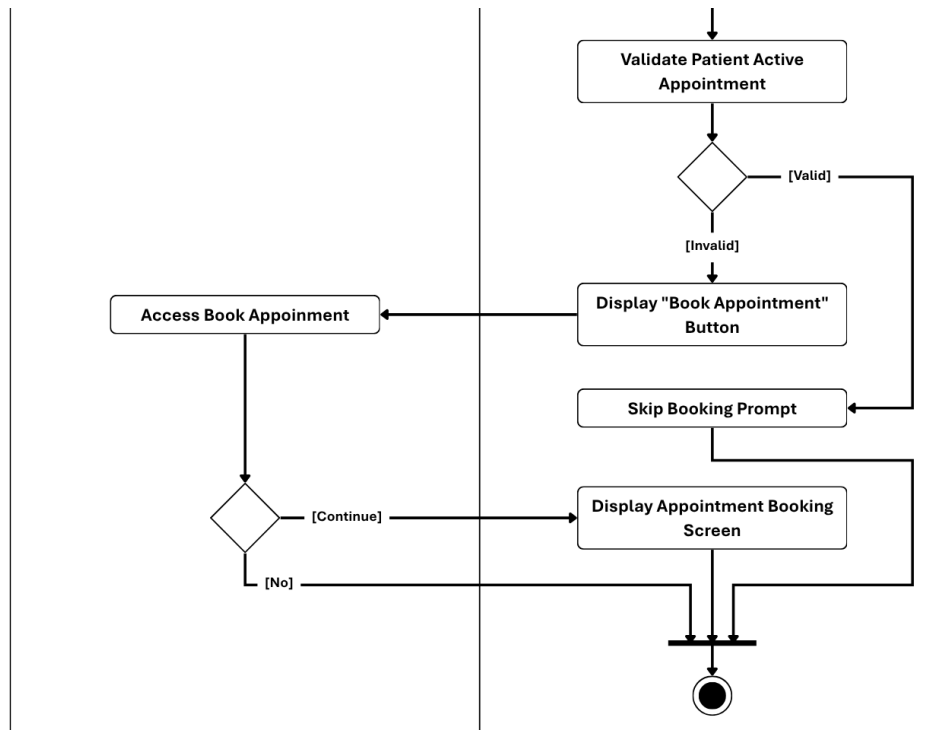


Figure 7. Recommended Visit Window Activity Diagram: Mobile

Figure 18 illustrates how the recommended visit feature generates a recommended follow-up window. The patient opens the recommended visit screen, enters oral health factors and quick logs, and the system checks the submitted data, appointment history, treatment records, and recorded next visit details before computing and displaying the recommended visit result.

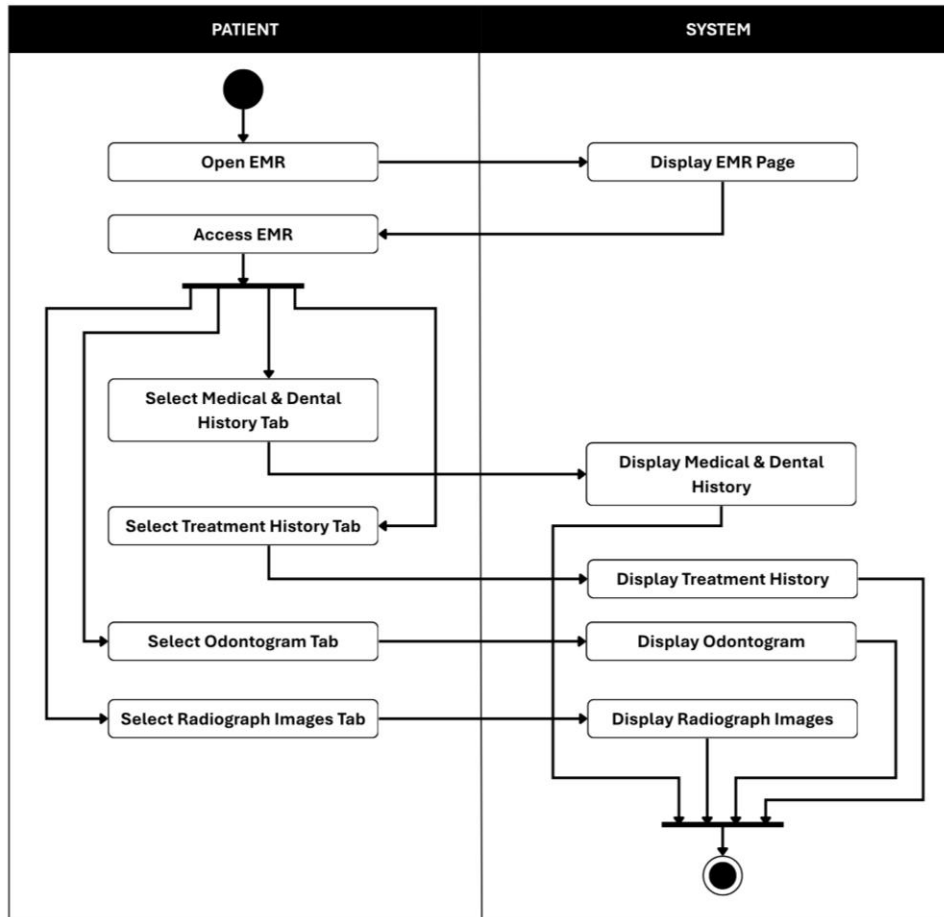


Figure 8. Electronic Medical Record Activity Diagram: Mobile

Figure 19 illustrates how the patient accesses the mobile EMR section. The patient opens the EMR page and selects the medical and dental history, treatment history, odontogram, and radiograph image tabs, while the system displays the corresponding read-only record section.

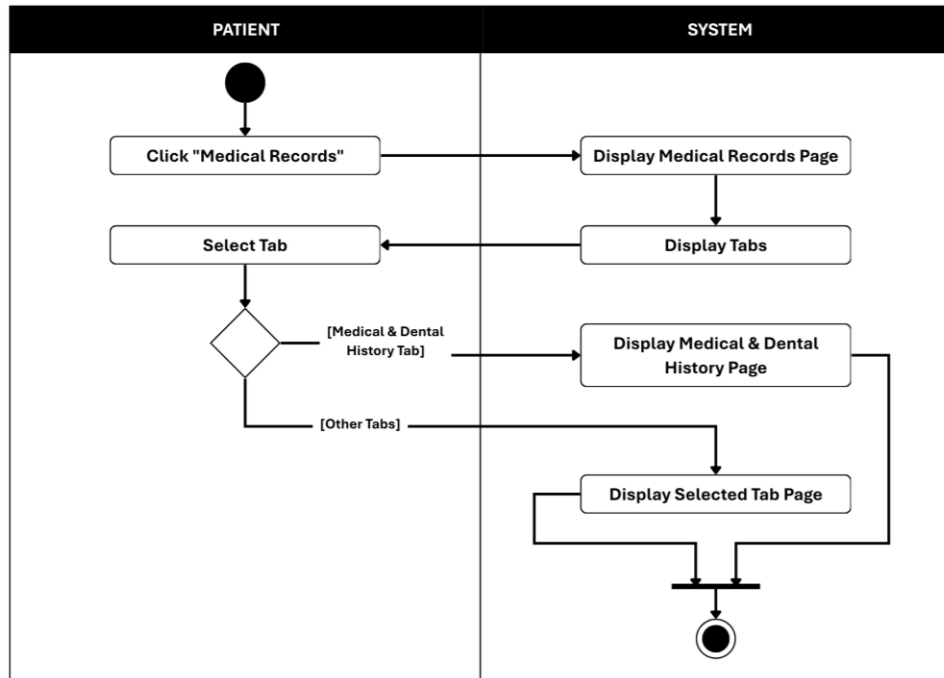


Figure 9. Medical and Dental History Activity Diagram: Mobile

Figure 20 illustrates how the patient views medical and dental history records in the mobile application. The patient opens Medical Records, selects the Medical and Dental History tab, and the system displays the appropriate history page or redirects the patient to another selected tab.

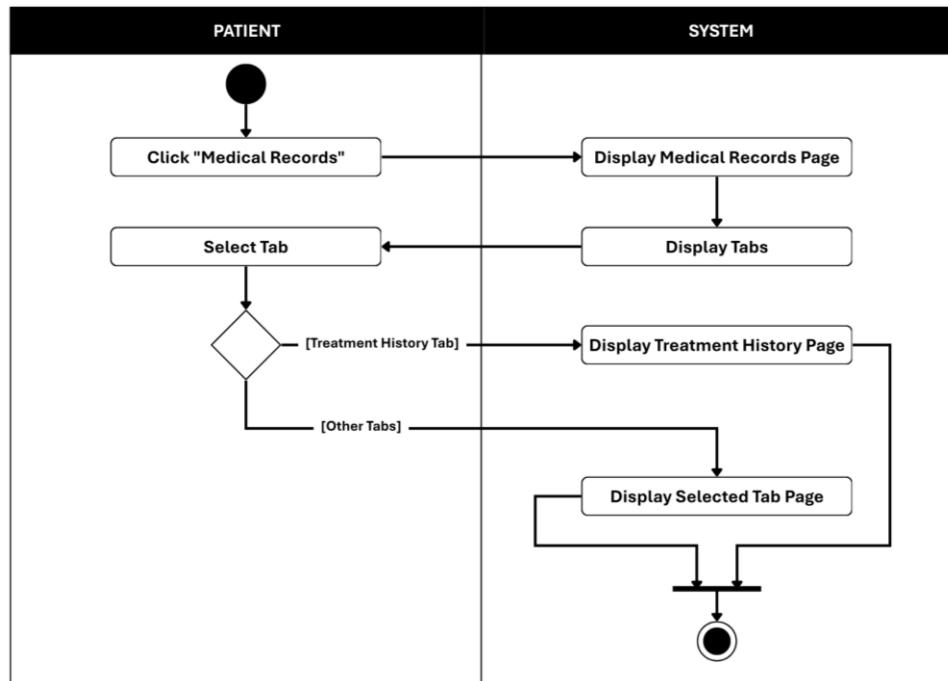


Figure 10. Treatment History Activity Diagram: Mobile

Figure 21 illustrates how the patient views treatment history records. The patient opens Medical Records, selects the Treatment History tab, and the system displays the treatment history page or navigates to another selected medical record tab.

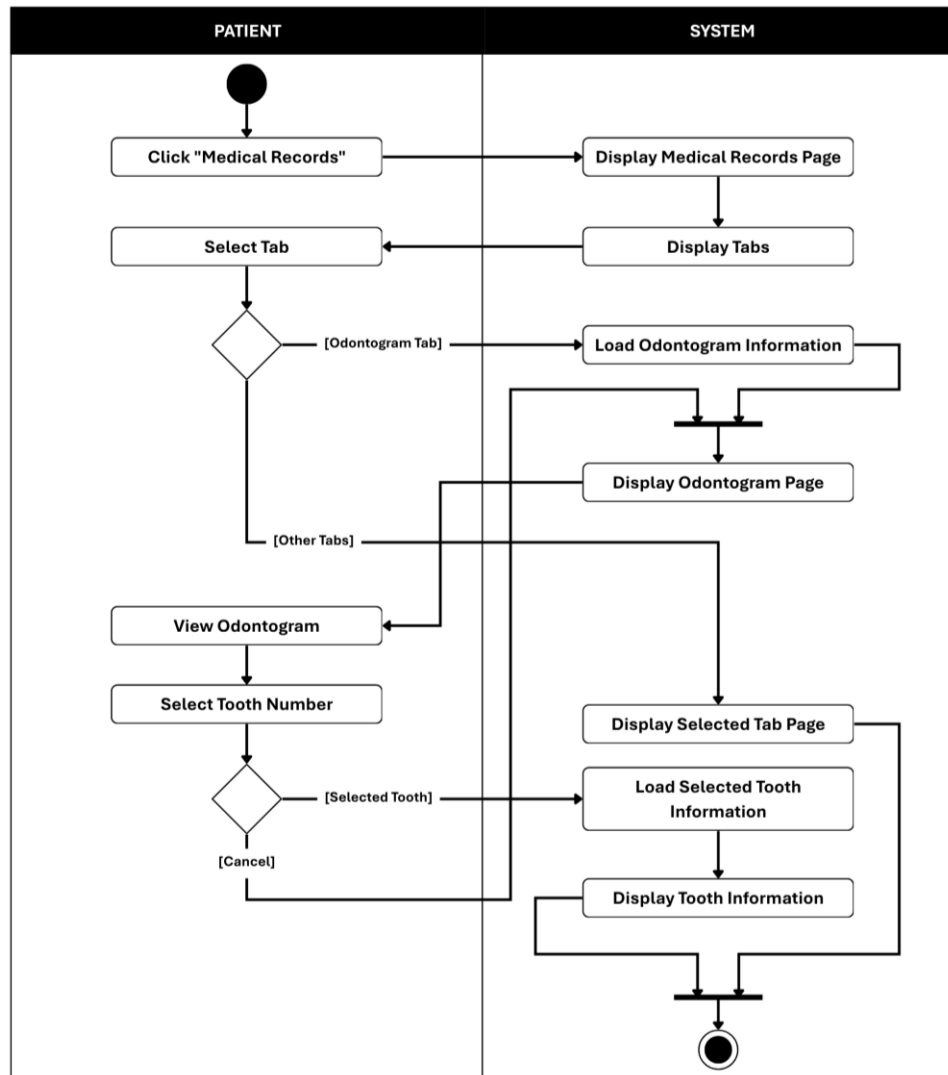


Figure 11. Odontogram Activity Diagram: Mobile

Figure 22 illustrates how the patient views odontogram information through the mobile EMR. The patient selects the Odontogram tab, views the odontogram page, and may select a tooth number, while the system loads and displays the selected tooth information.

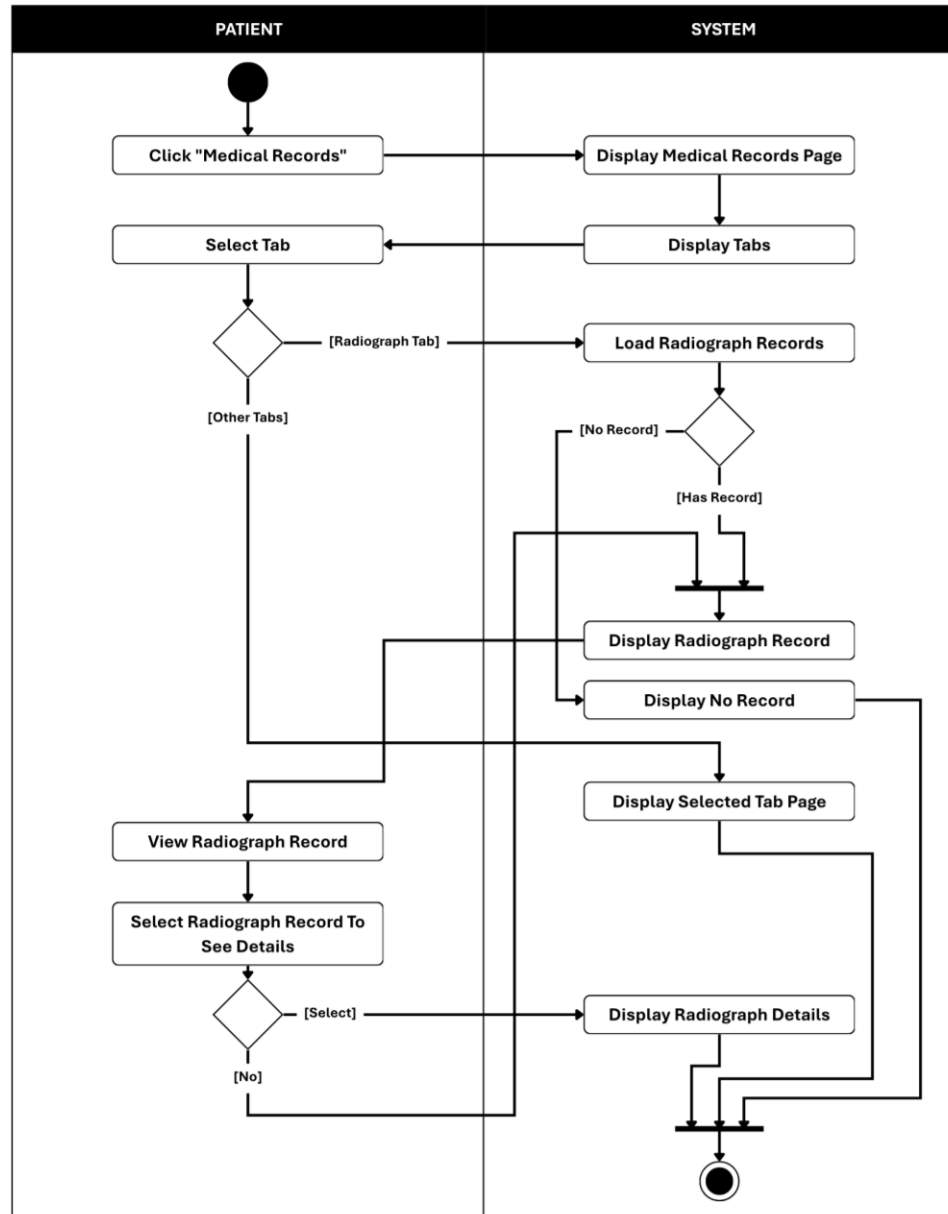


Figure 12. Radiograph Activity Diagram: Mobile

Figure 23 illustrates how the patient views radiograph records in the mobile application. The patient selects the Radiograph tab, and the system loads available radiograph records, displays either the record list or a no-record message, and shows radiograph details when the patient selects a record.

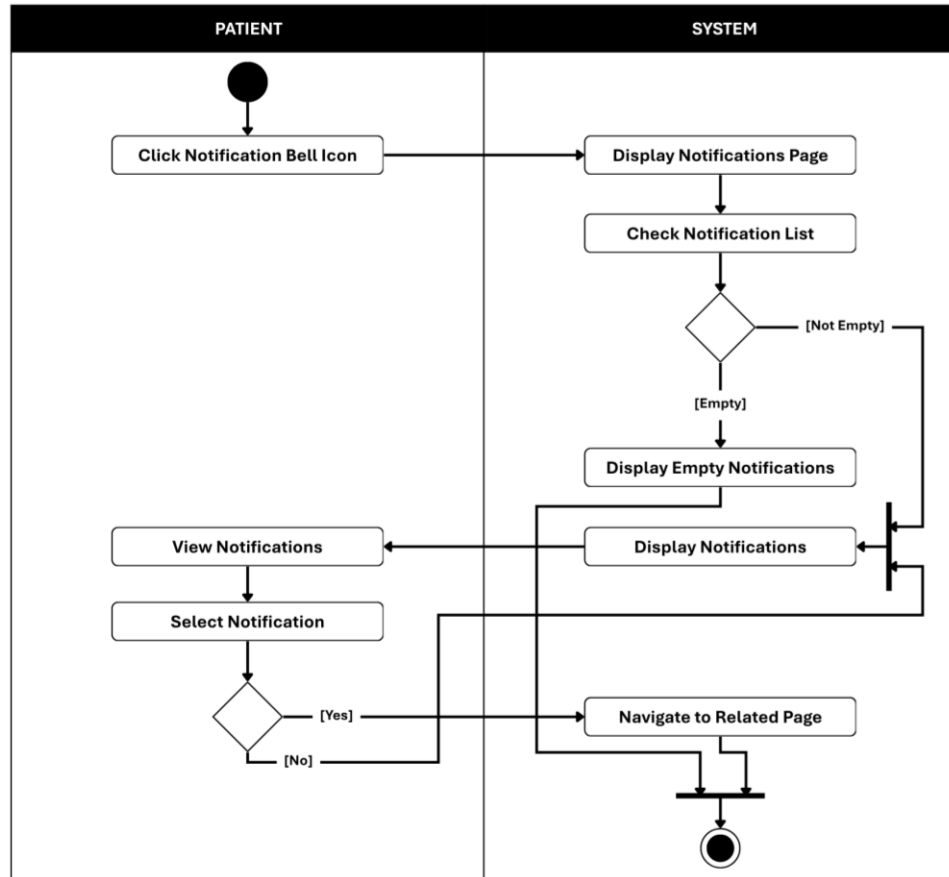


Figure 13. Notifications Activity Diagram: Mobile

Figure 24 illustrates how mobile notifications are viewed. The patient taps the notification bell icon, and the system displays the notifications page, checks whether notifications are available, shows an empty state or notification list, and redirects the patient to a related page when a notification is selected.

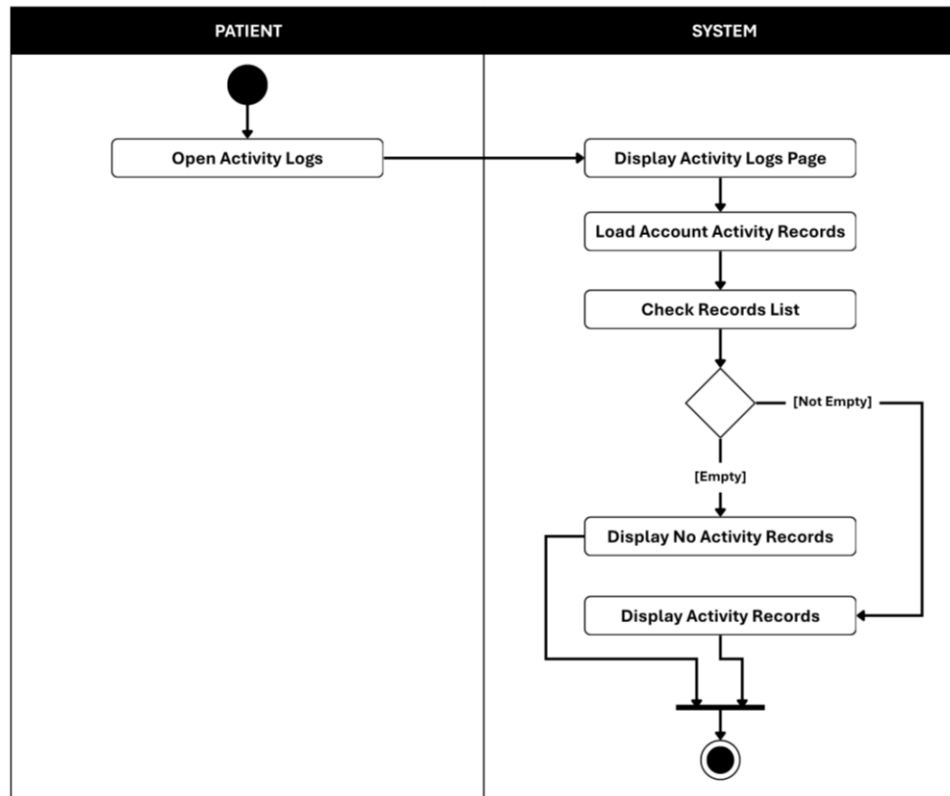
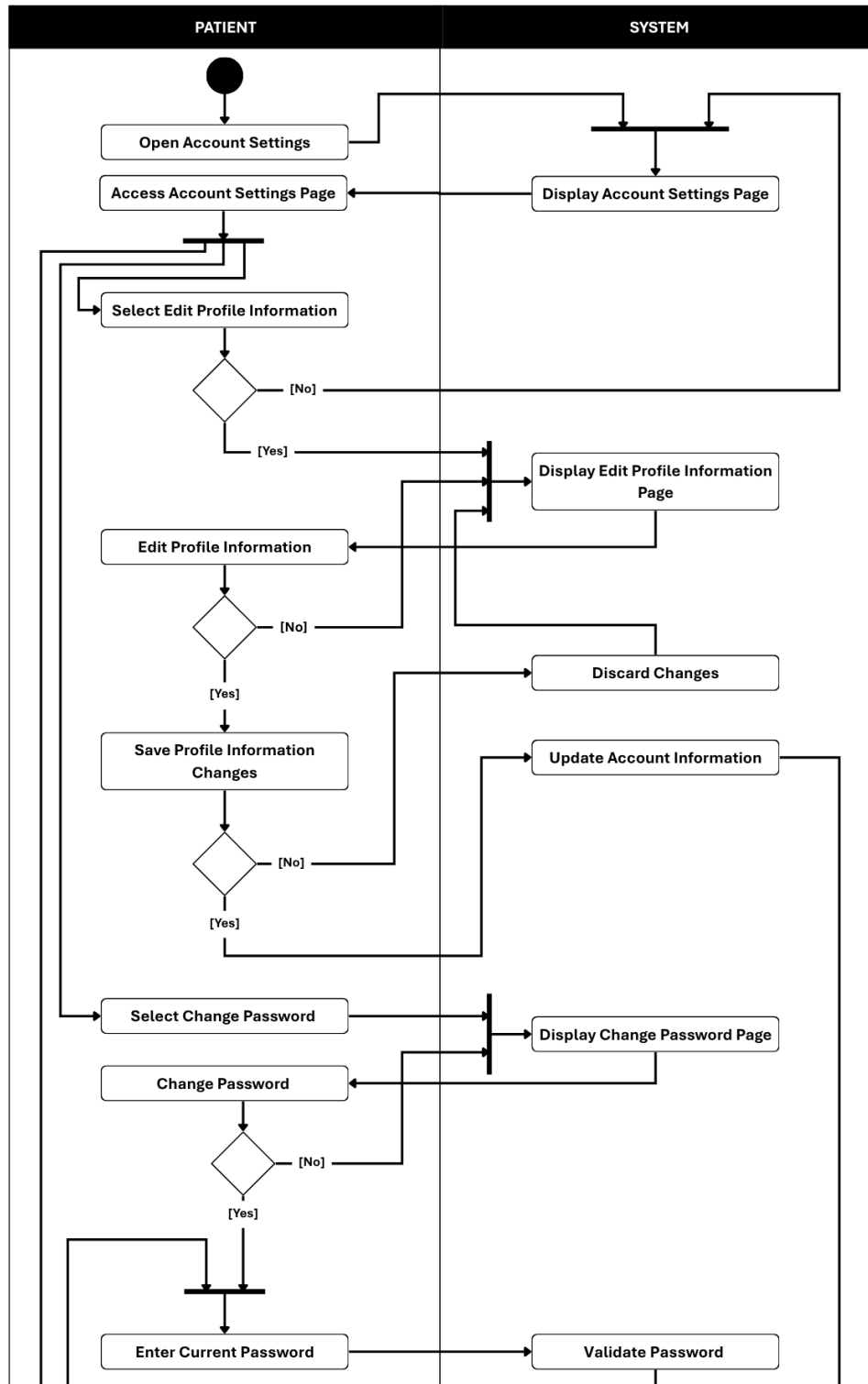


Figure 14. Activity Logs Activity Diagram: Mobile

Figure 25 illustrates how the system displays patient activity logs. The patient opens Activity Logs, and the system loads account activity records, checks whether records exist, and displays either the activity history or a no-records message.



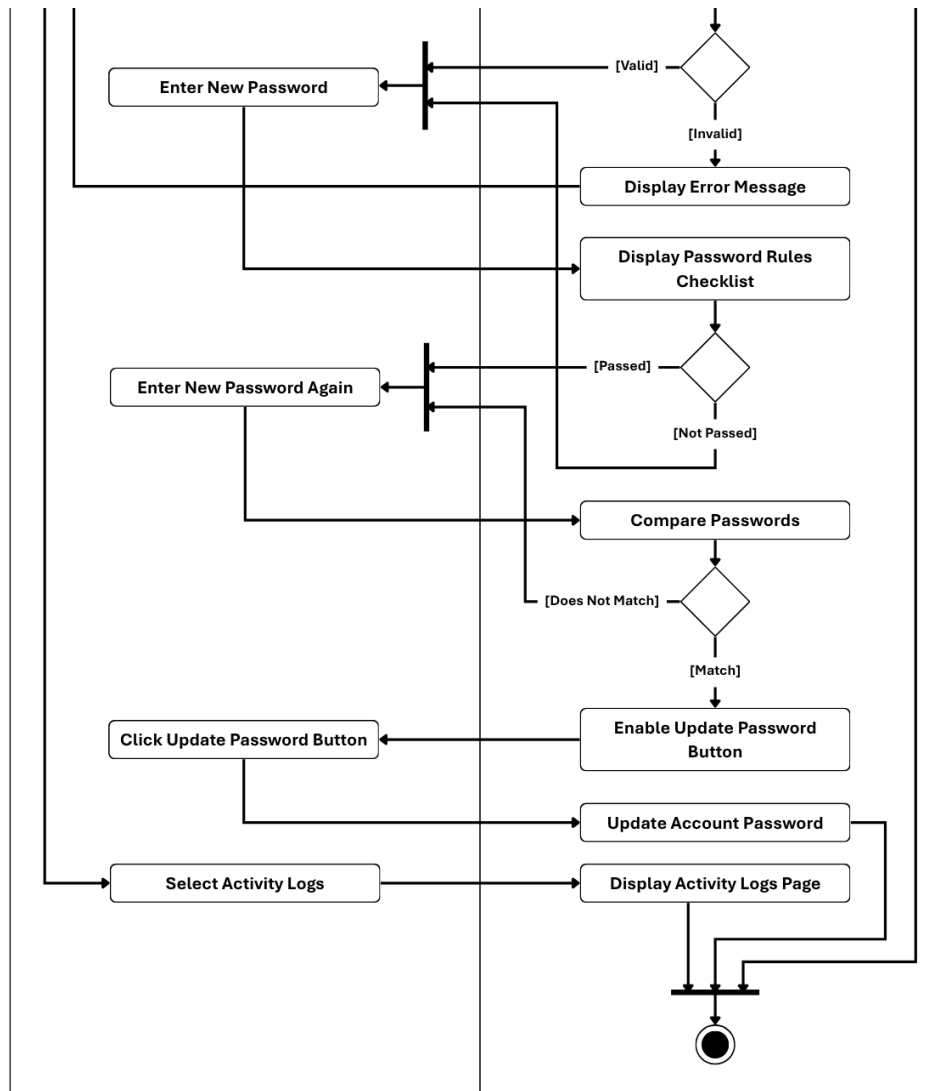


Figure 15. Account Settings Activity Diagram: Mobile

Figure 26 illustrates the account settings process in the mobile application. The patient opens Account Settings and may edit profile information, change the account password, or open activity logs. The system validates changes, displays errors or password rules when needed, saves valid updates, and updates the account information.

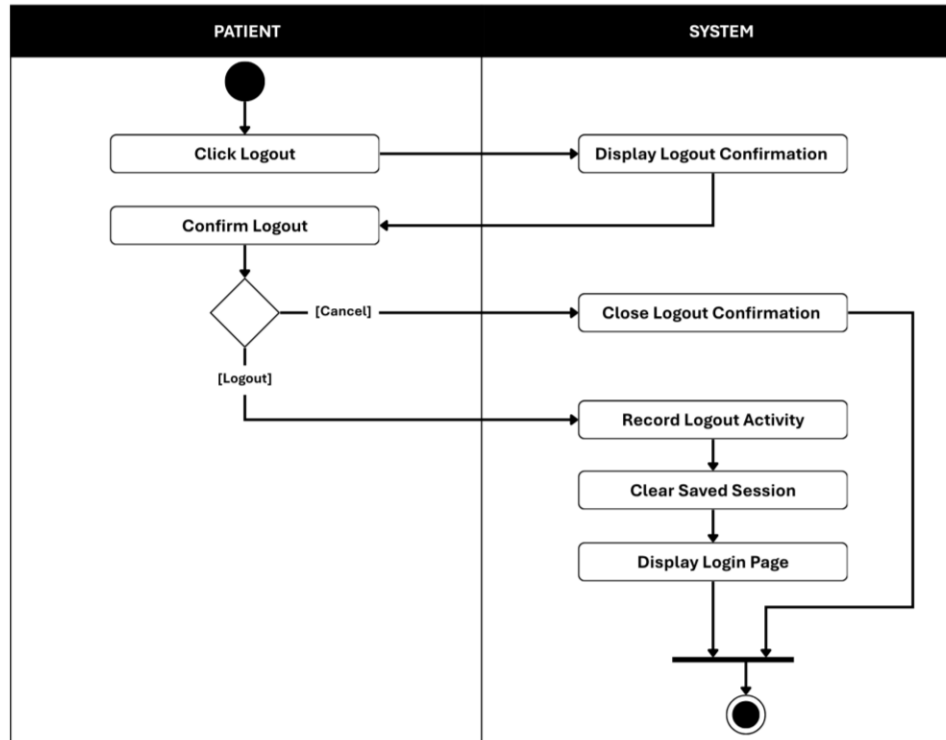


Figure 16. Logout Activity Diagram: Mobile

Figure 27 illustrates the mobile logout process. The patient clicks Logout and confirms the action; the system closes the confirmation dialog, records the logout activity, clears the active session, and returns the patient to the login page.

NGITIFY WEB ACTIVITY DIAGRAM

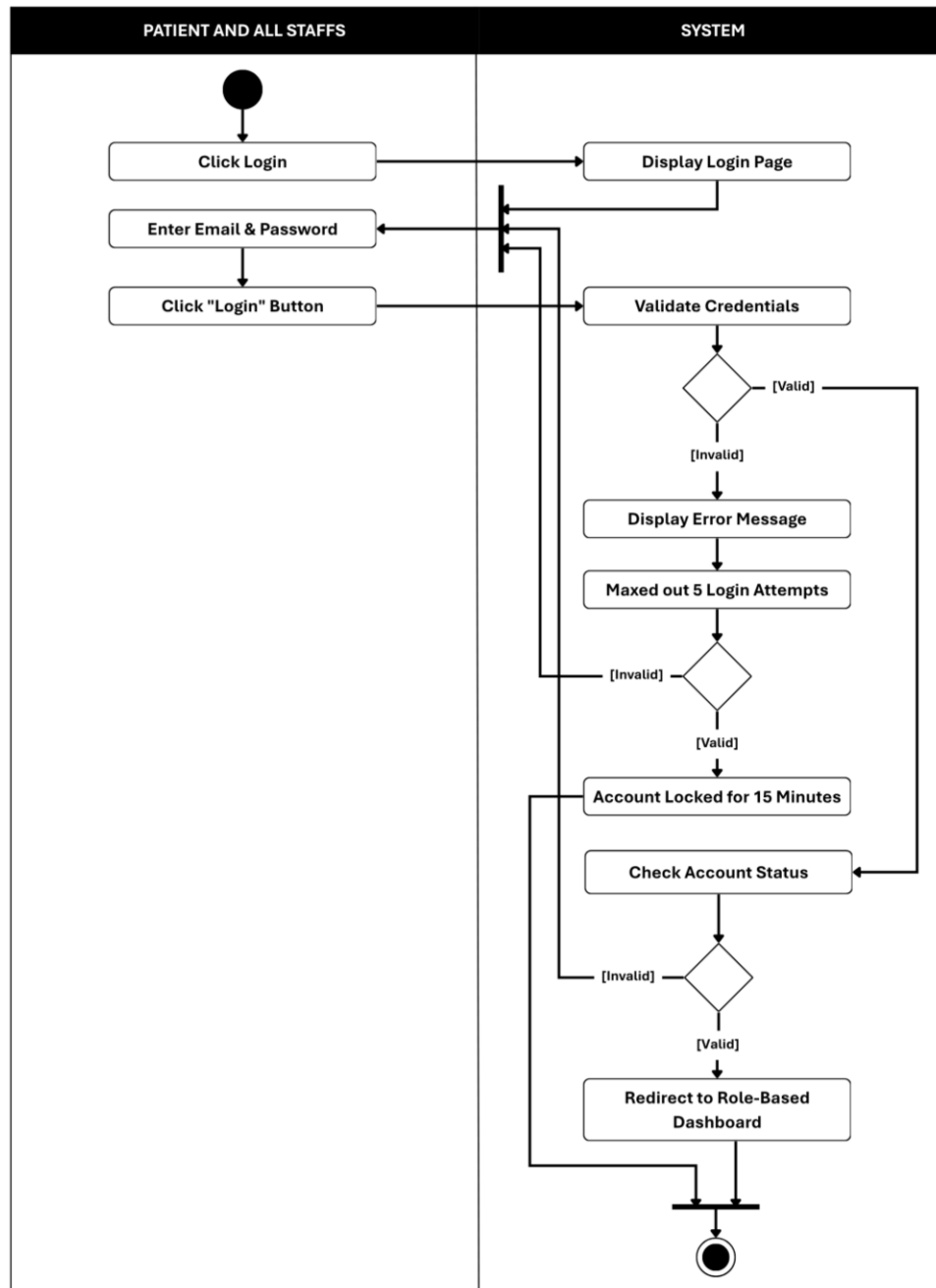


Figure 17. Login Activity Diagram: Web

Figure 28 illustrates the web login process for patients and staff users. The user enters an email address and password, and the system validates the credentials, checks failed login attempts and account status, displays errors as needed, locks the account after repeated failed attempts, and redirects the user to the appropriate role-based dashboard after a successful login.

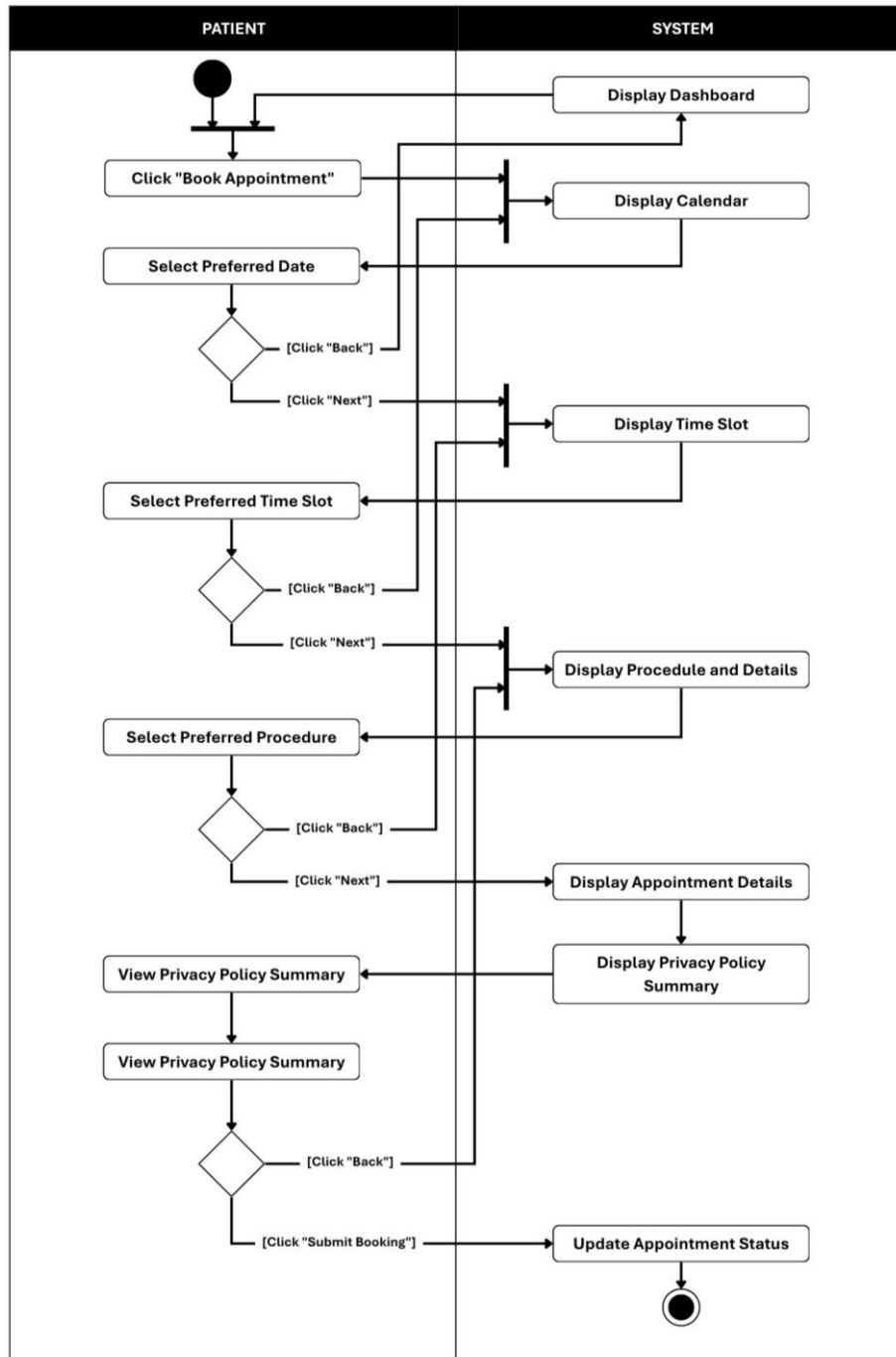
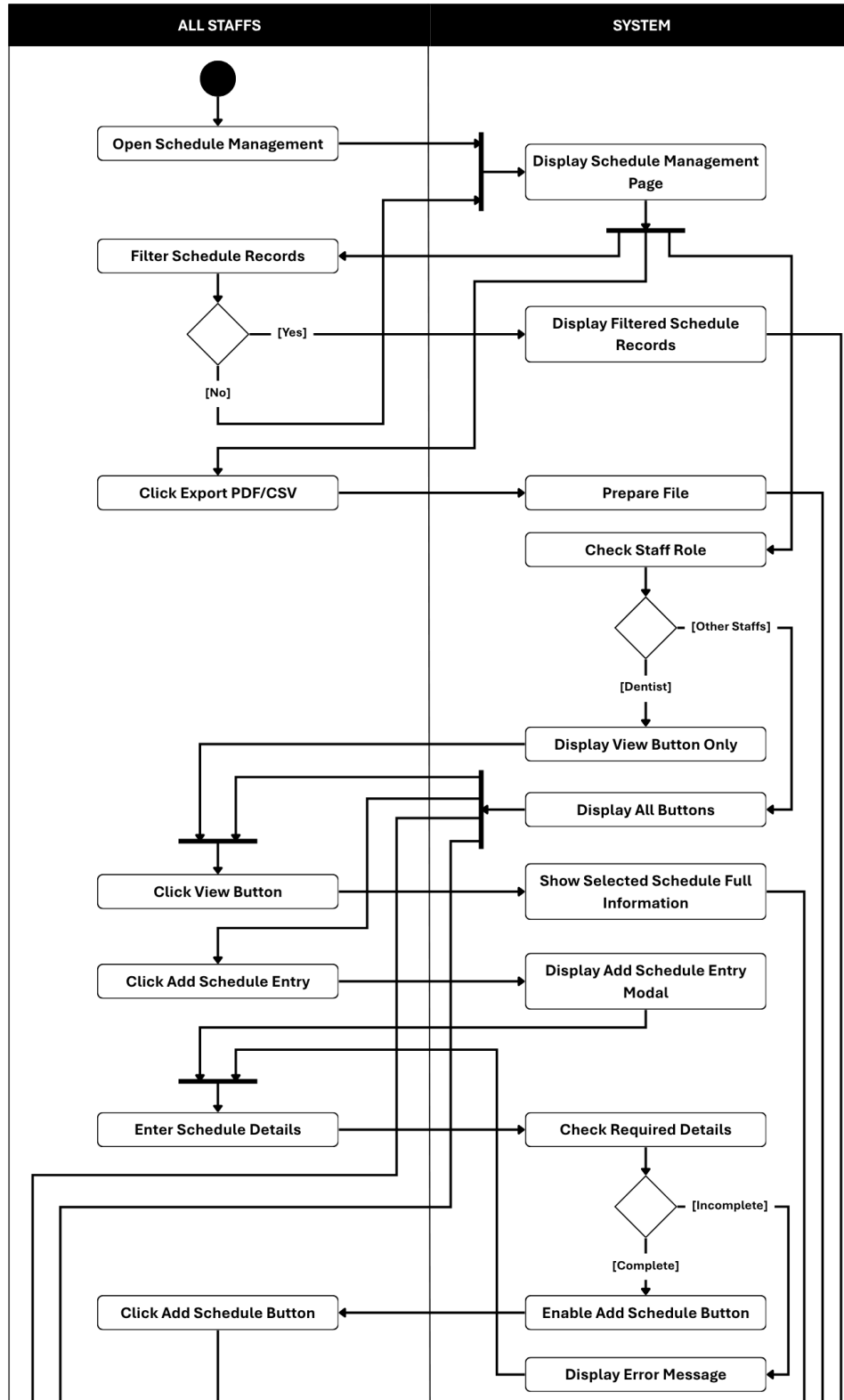


Figure 18. Appointment Scheduling Activity Diagram: Web

Figure 29 illustrates the web appointment scheduling process for a patient. The patient selects a preferred date, time slot, and procedure, reviews the appointment and privacy policy summary, and submits the request. The system displays the calendar, validates the selections, shows the appointment details, and updates the appointment status after booking.



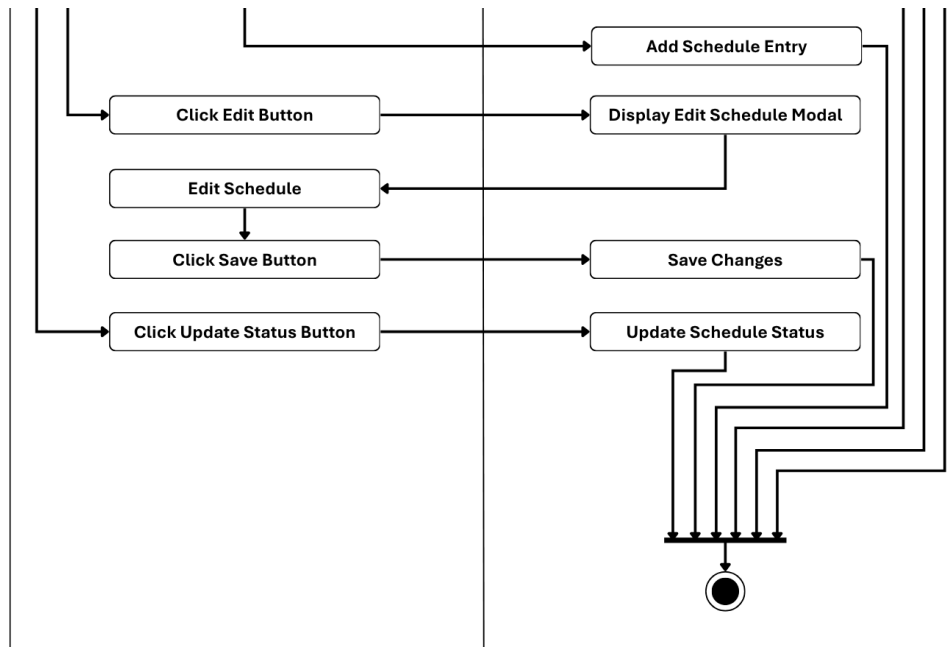


Figure 19. Schedule Management Activity Diagram: Web

Figure 30 illustrates how staff manage clinic schedules in the web system. The user opens Schedule Management, filters schedule records, changes the schedule view, exports records, adds or edits schedule entries, and deletes or updates schedule status. At the same time, the system validates inputs, displays the proper schedule list, and saves the changes.

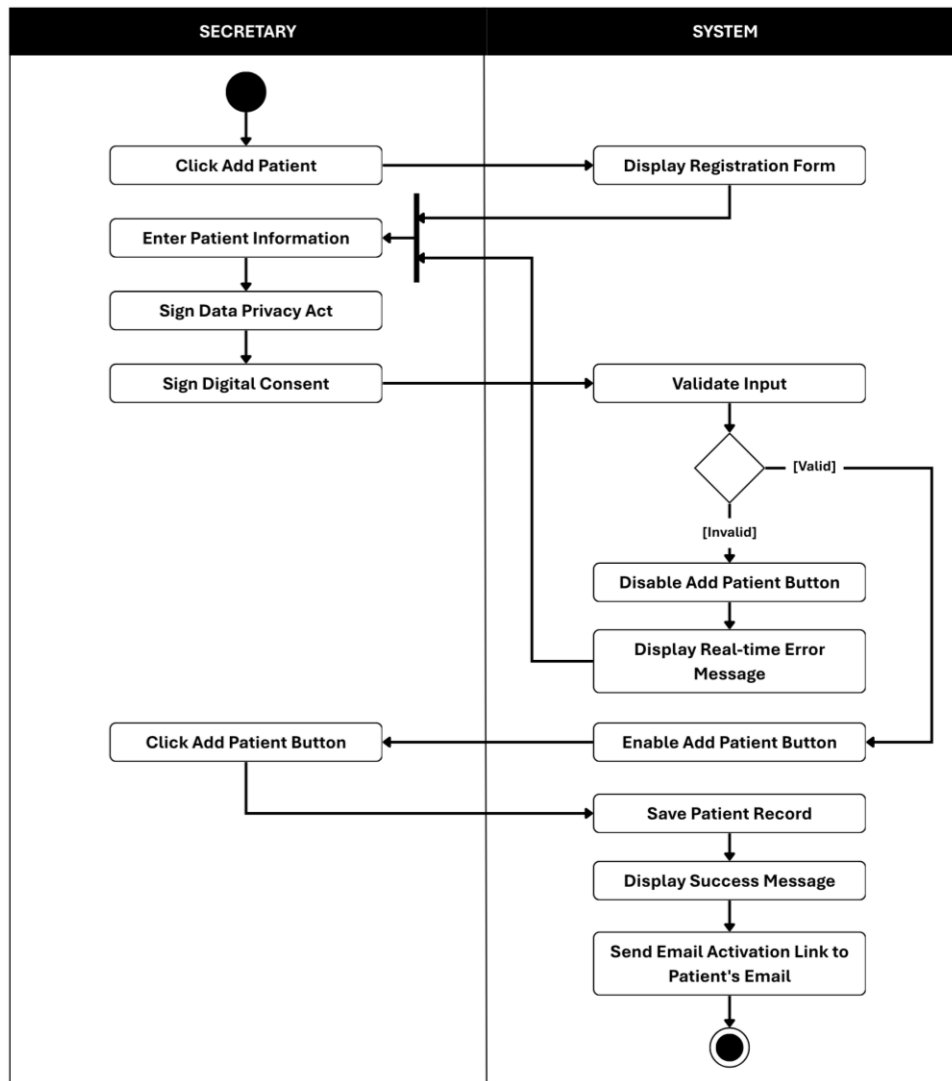
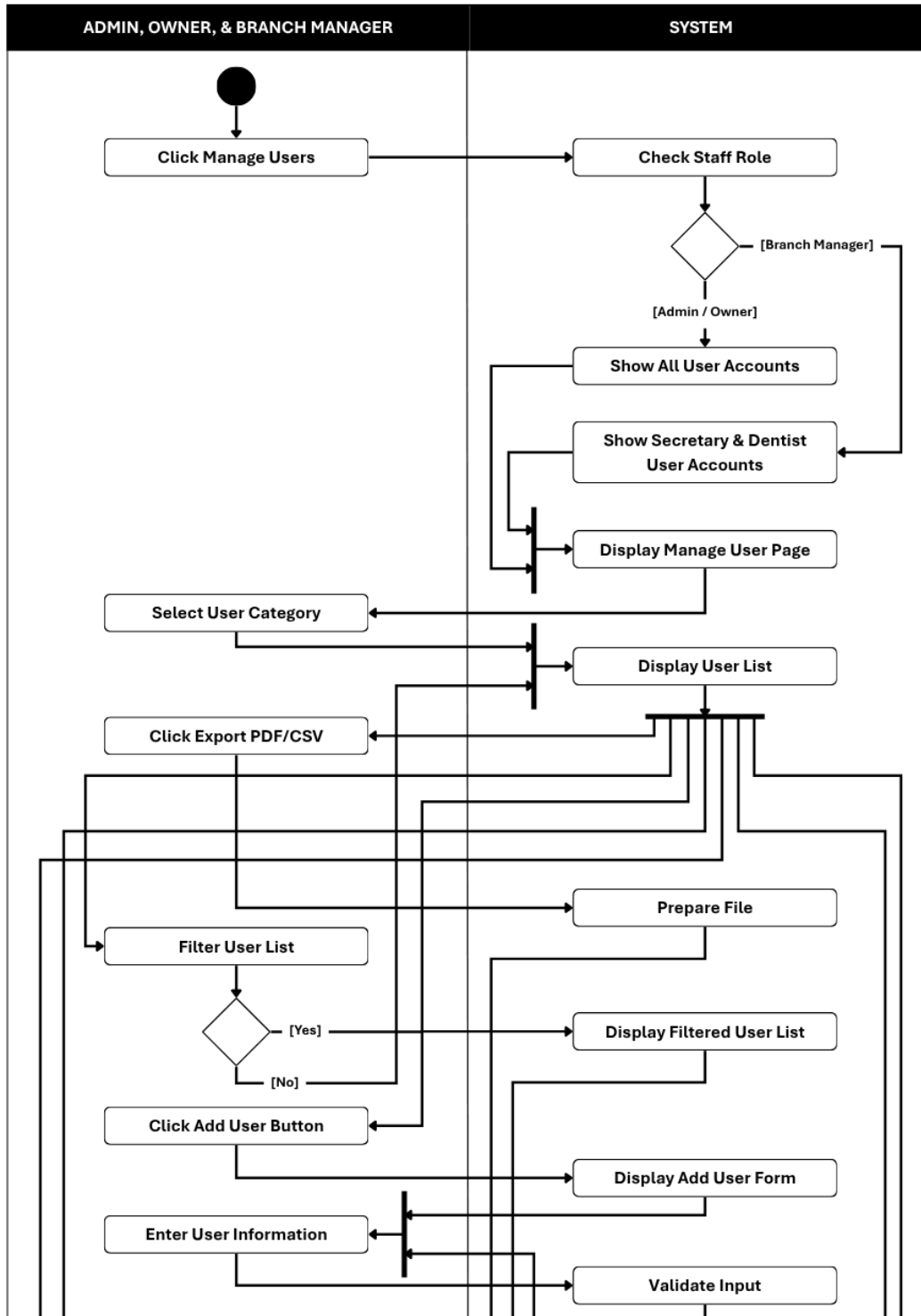
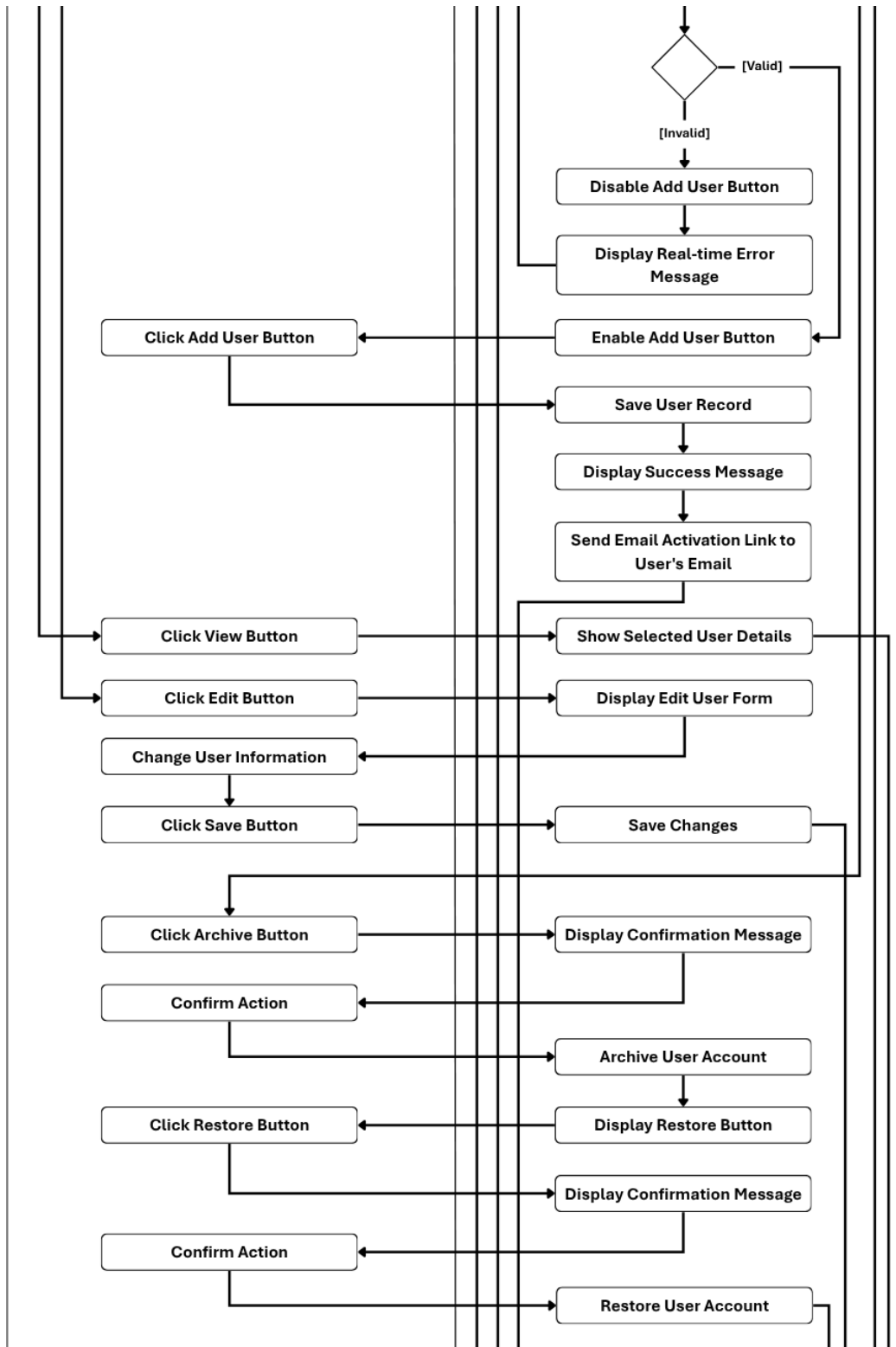


Figure 20. Patient Registration Activity Diagram: Web

Figure 31 illustrates the patient registration process the secretary handles. The secretary opens the registration form, enters patient information, confirms data privacy and digital consent, and submits the record. The system validates the input, displays real-time errors for invalid entries, saves the patient record, displays a success message, and sends an activation link to the patient's email.





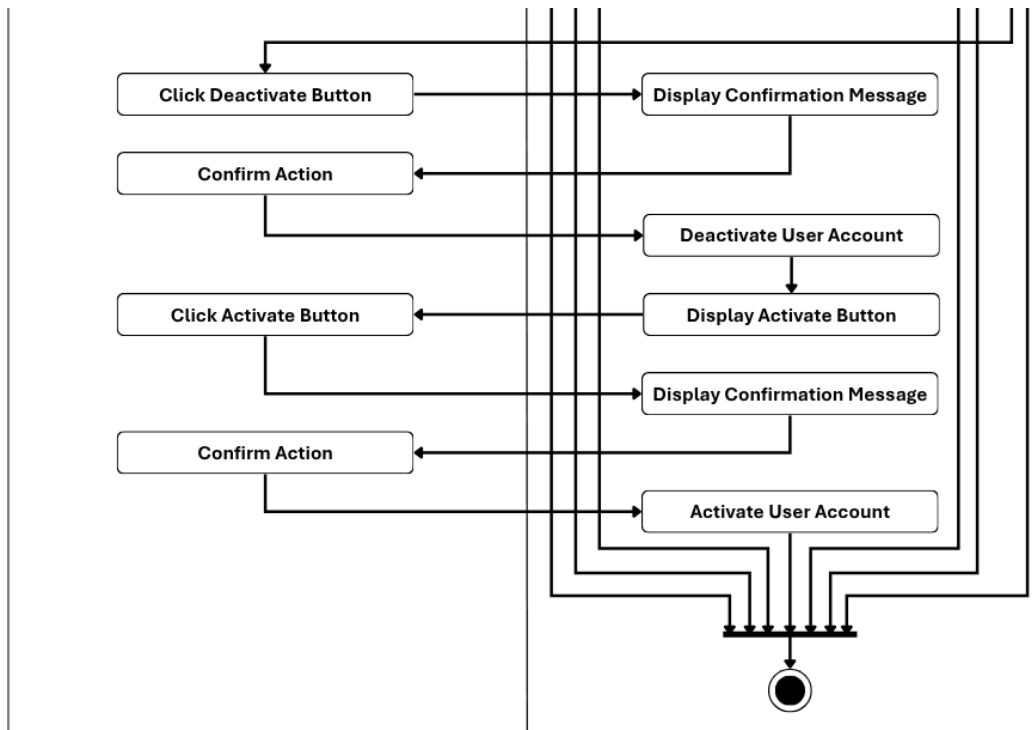


Figure 21. User Account Management Activity Diagram: Web

Figure 32 illustrates how authorized users manage system accounts. The system checks the staff role, displays the appropriate user account list, supports filtering and export, and allows users to add, view, edit, archive, restore, deactivate, or activate accounts, with validation and confirmation steps.

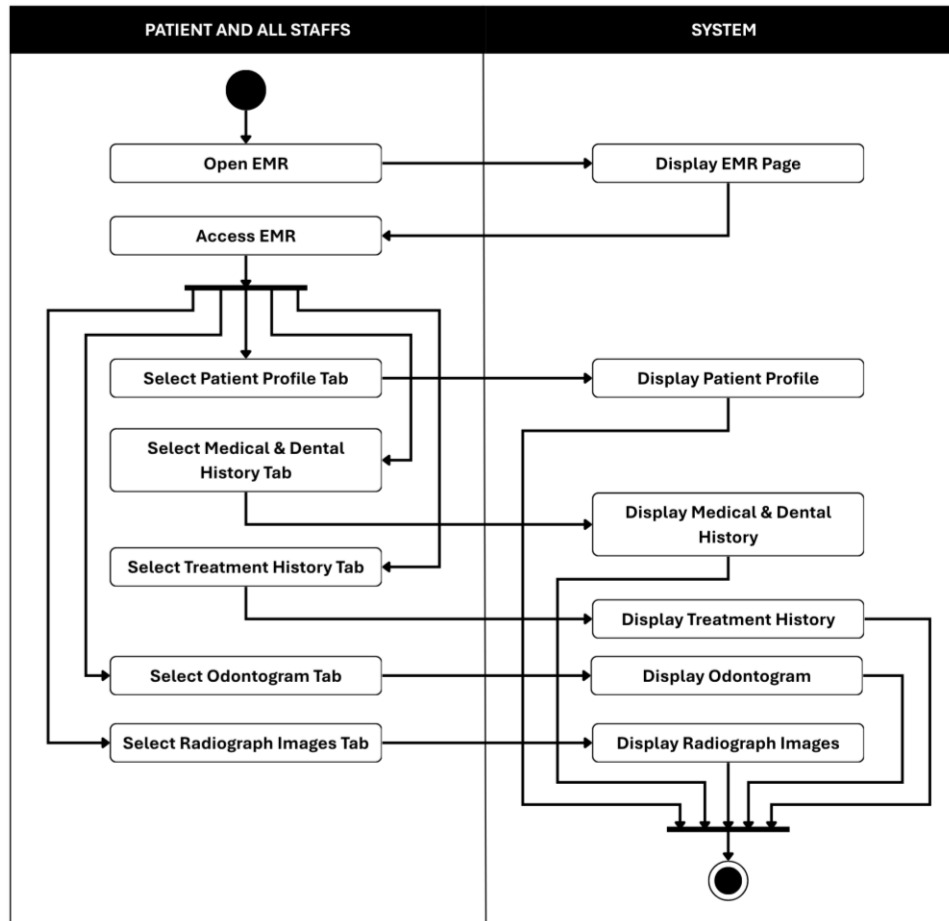


Figure 22. EMR Activity Diagram: Web

Figure 33 illustrates the web EMR access process. The user opens the EMR page and selects among patient profile, medical and dental history, treatment history, odontogram, and radiograph image tabs, while the system displays the selected EMR section.

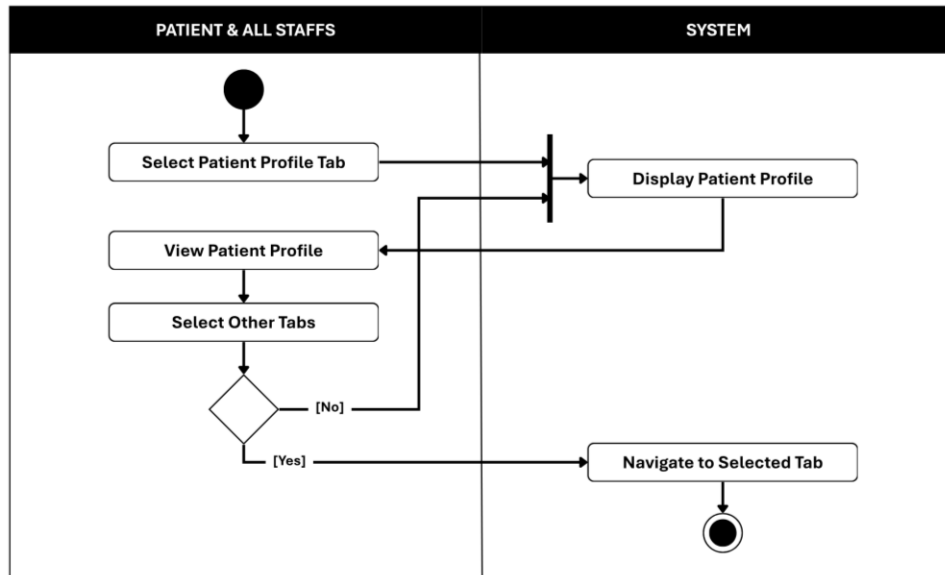


Figure 23. Patient Profile Activity Diagram: Web

Figure 34 shows how the patient profile tab appears in the EMR. The user selects the Patient Profile tab, and the system displays the patient's profile information or redirects the user to another selected tab.

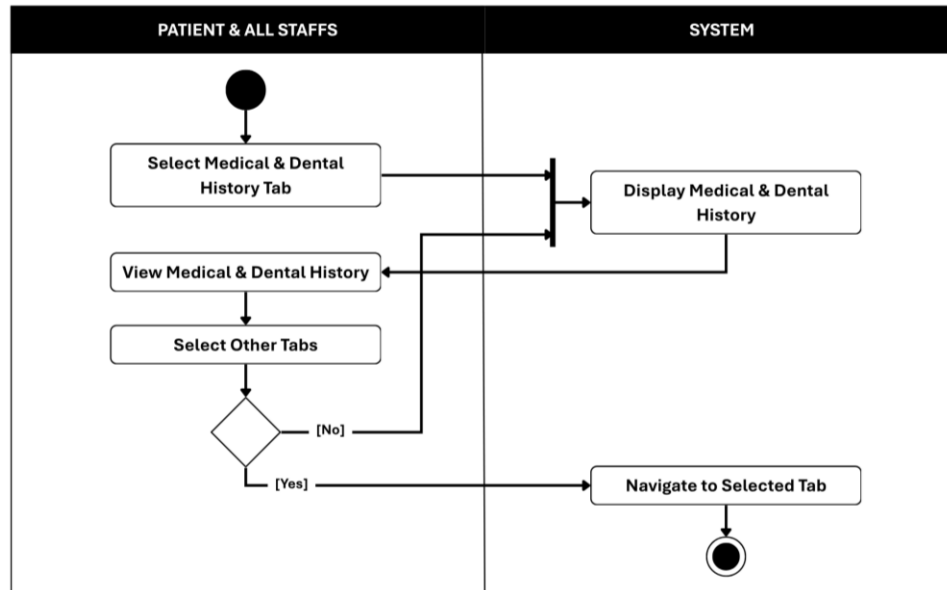


Figure 24. Medical and Dental History Activity Diagram: Web

Figure 35 illustrates how users access medical and dental history records in the web EMR. The user selects the Medical and Dental History tab, views the displayed history, and may navigate to another EMR tab.

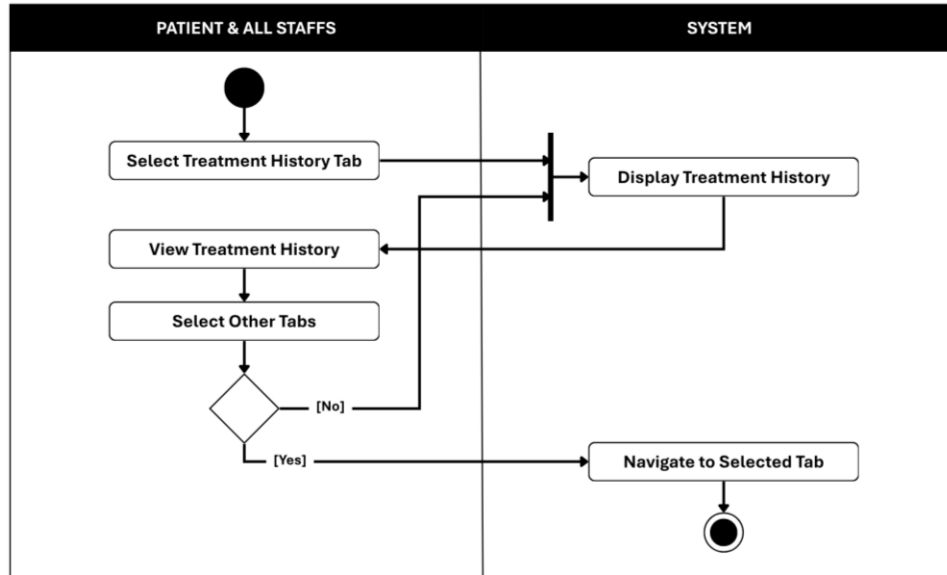


Figure 25. Treatment History Activity Diagram: Web

Figure 36 illustrates how the web EMR displays treatment history records. The user selects the Treatment History tab, and the system displays the treatment history or redirects the user to another selected tab.

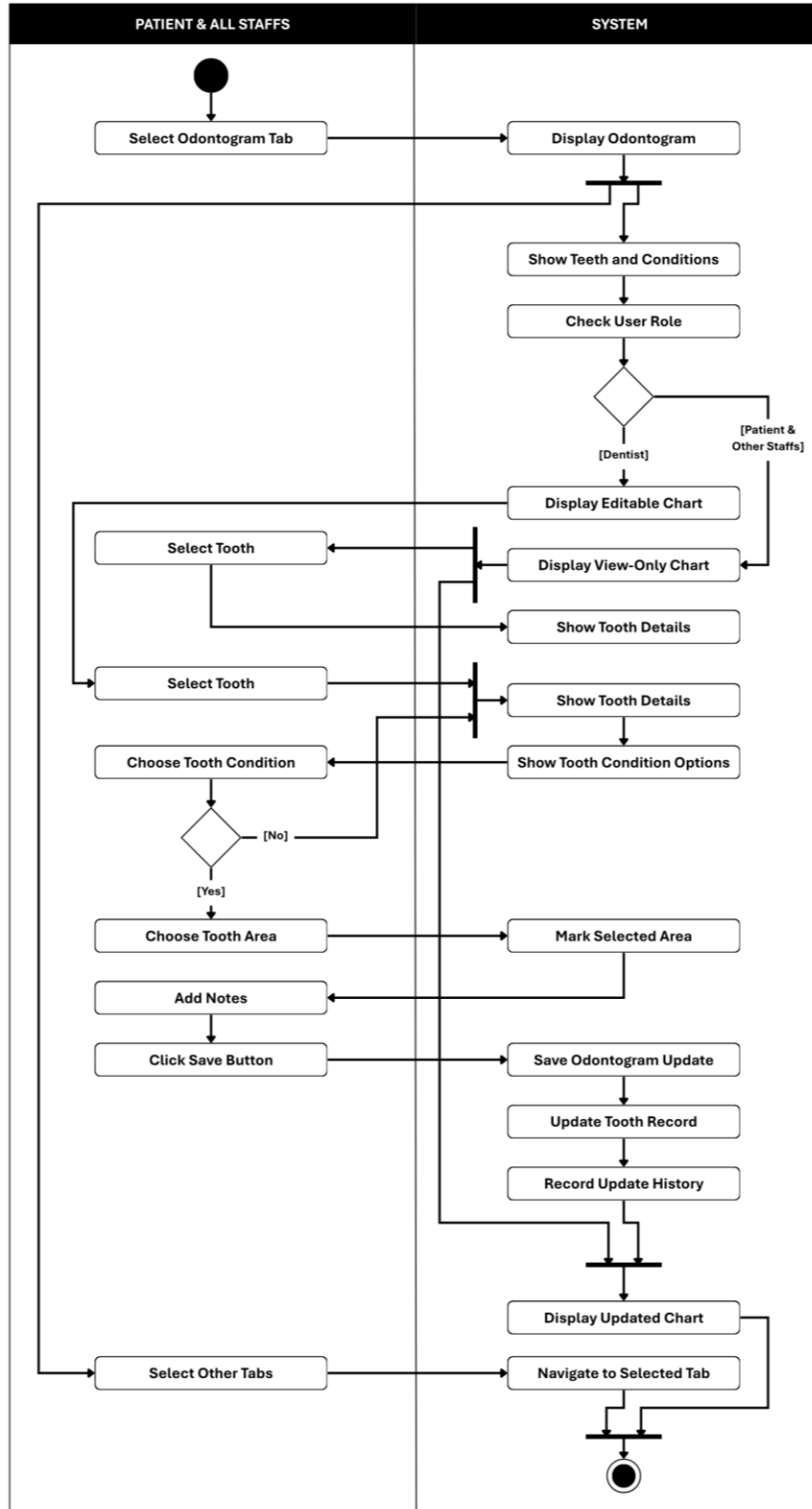


Figure 26. Odontogram Activity Diagram: Web

Figure 37 illustrates how the system views and updates odontogram information. The system displays the odontogram, checks the user role, shows static or view-only details when appropriate, and allows authorized users to select teeth, choose tooth conditions and areas, add notes, save updates, record update history, and display the updated chart.

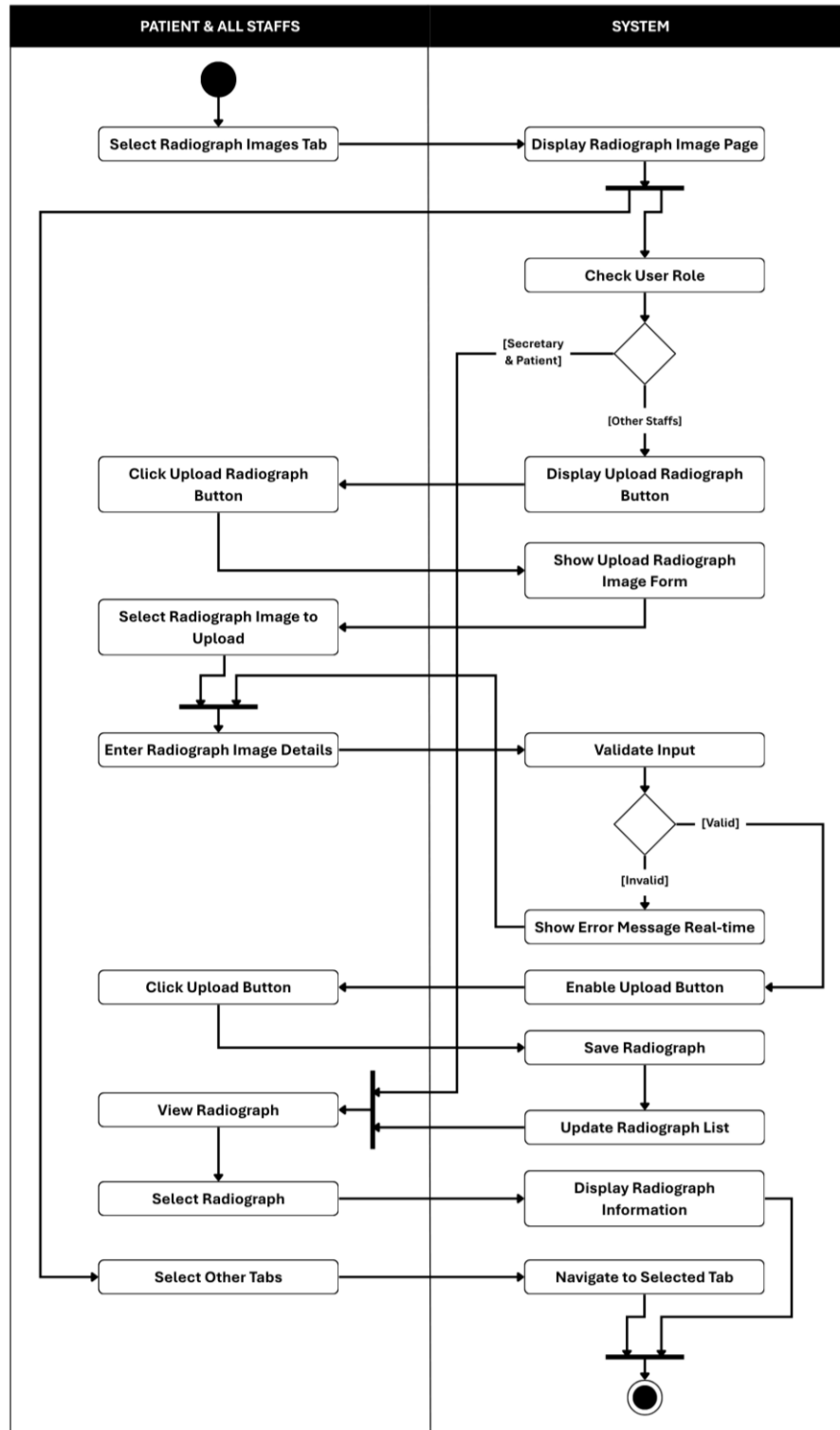


Figure 27. Radiograph Images Activity Diagram: Web

Figure 38 illustrates how the web EMR manages radiograph images. The system displays the radiograph page, checks the user role, allows authorized users to upload radiograph images with

validated details, saves the record, updates the radiograph list, and displays selected radiograph information.

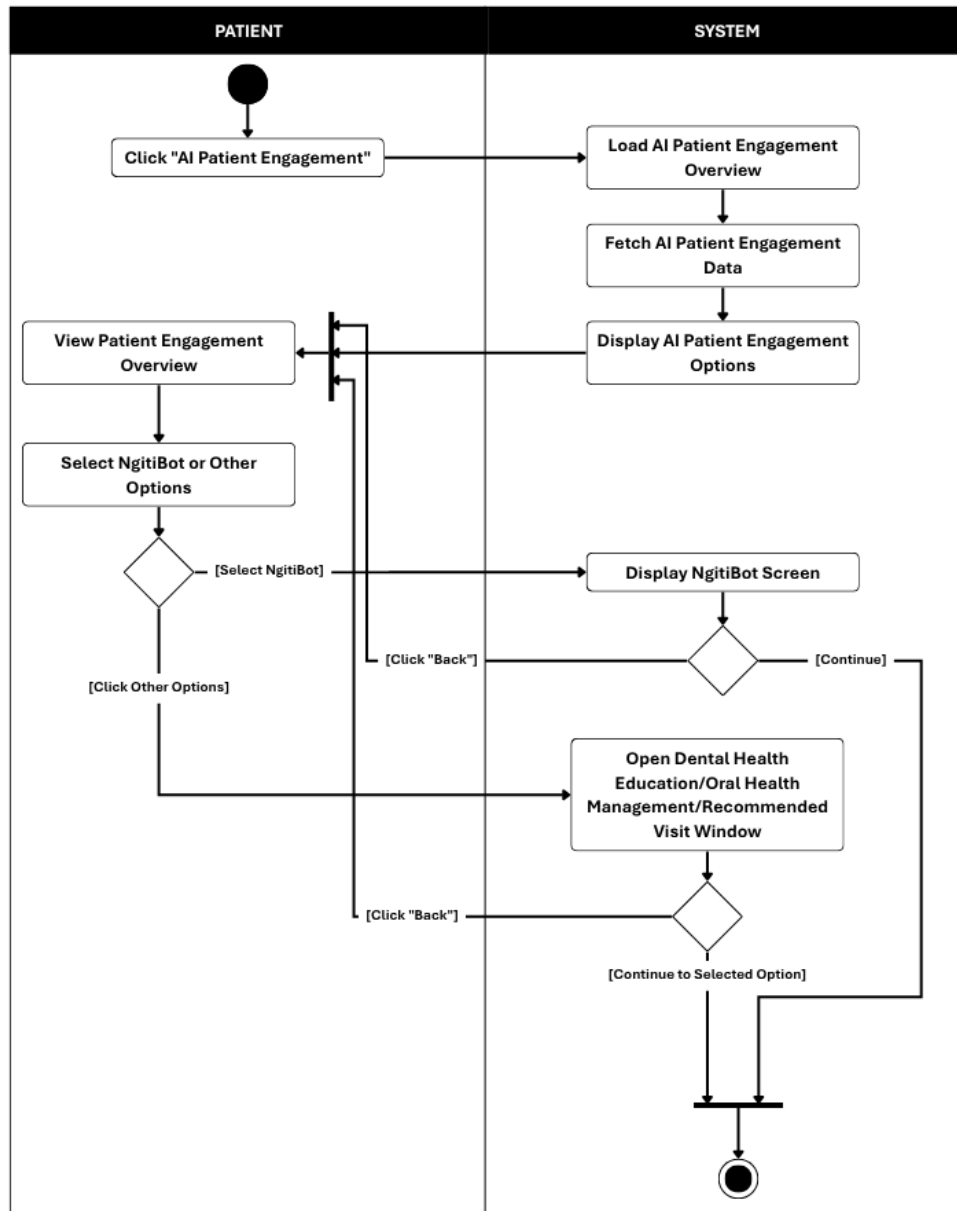


Figure 28. AI Patient Engagement Activity Diagram: Web

Figure 39 illustrates the patient engagement process on the web. The patient opens AI Patient Engagement, and the system loads engagement data, displays available options, and routes the patient to NgitiBot, dental health education, oral health management, or recommended visit features.

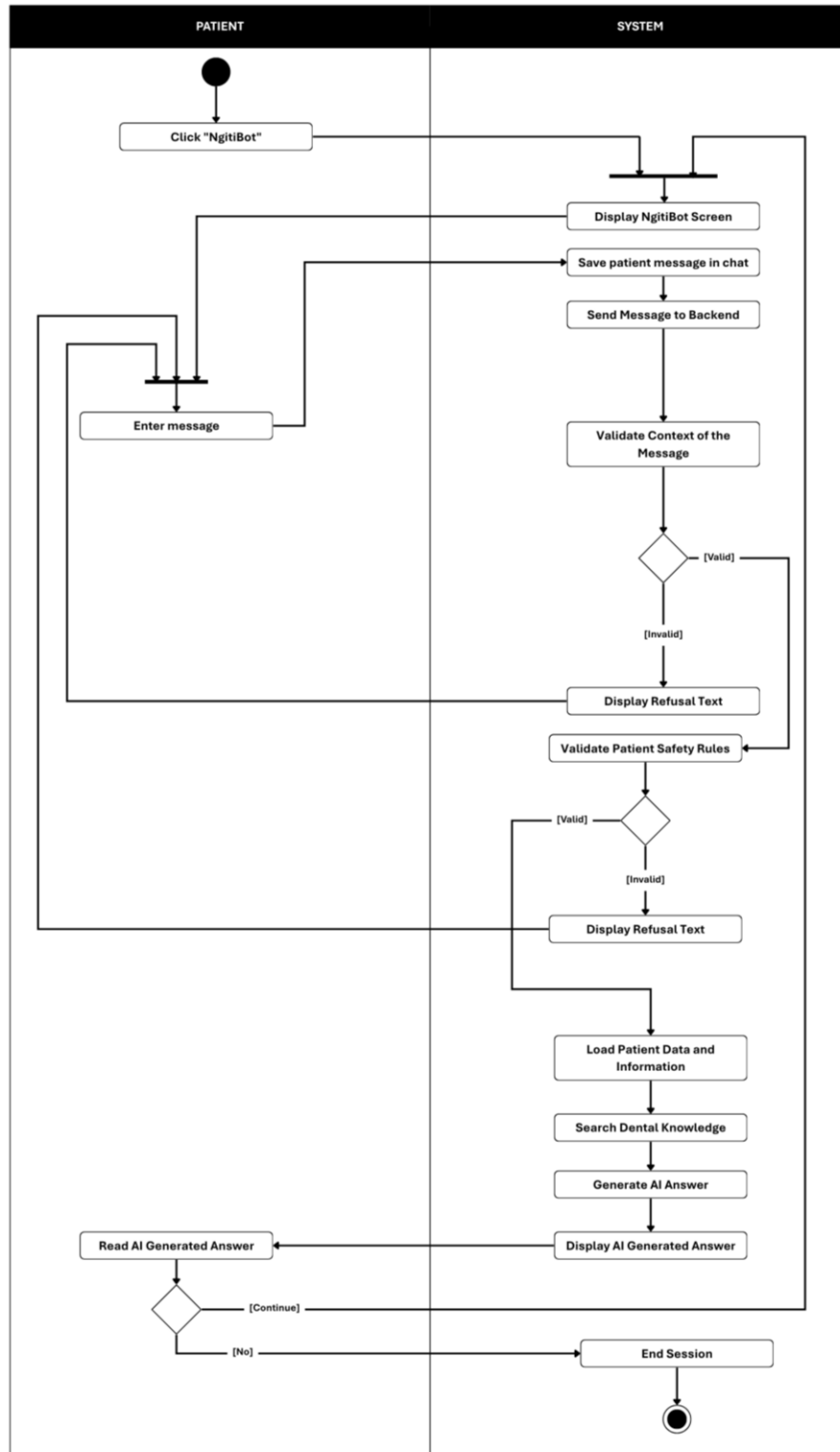


Figure 29. NgitiBot Activity Diagram: Web

Figure 40 illustrates the web NgitiBot conversation process. The patient submits a message, and the system saves it, sends it to the backend, validates the content and safety rules, retrieves relevant patient and dental knowledge, generates an AI answer, and displays the response or a refusal message.

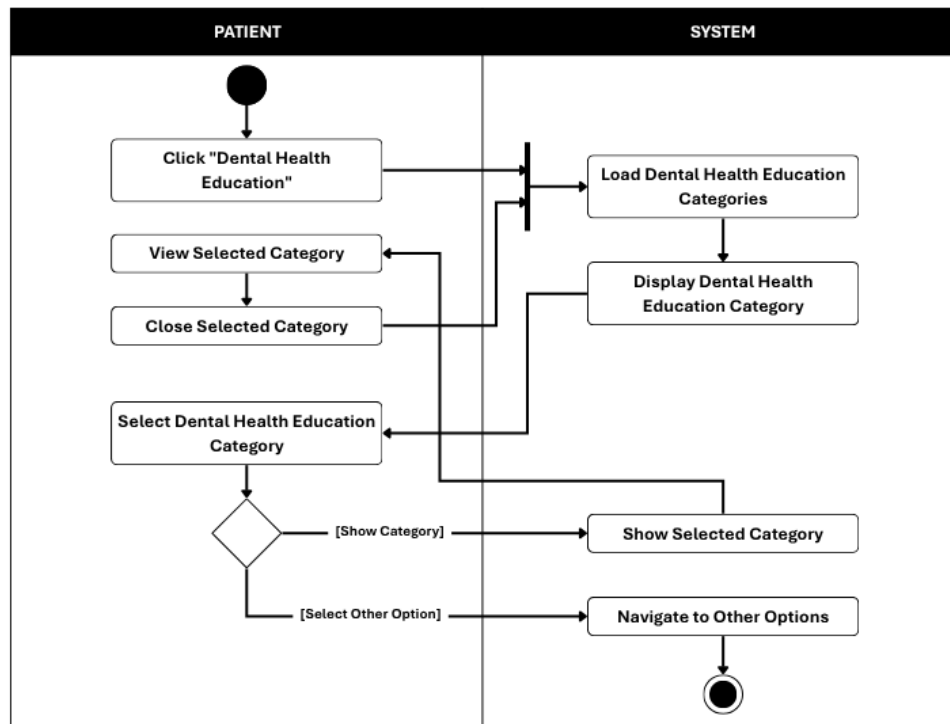


Figure 30. Dental Health Education Activity Diagram: Web

Figure 41 illustrates how the patient accesses dental health education content online. The patient opens Dental Health Education, views available categories, selects a category, and the system displays the selected educational content or navigates to another option.

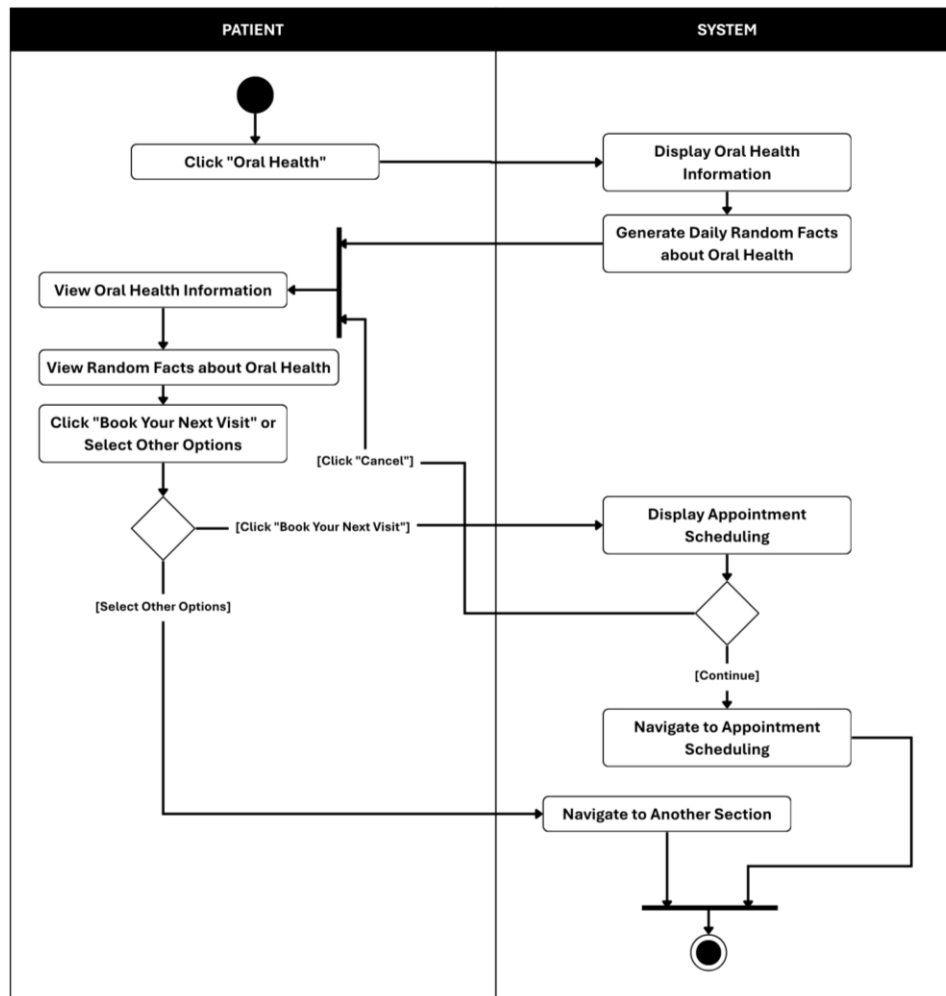
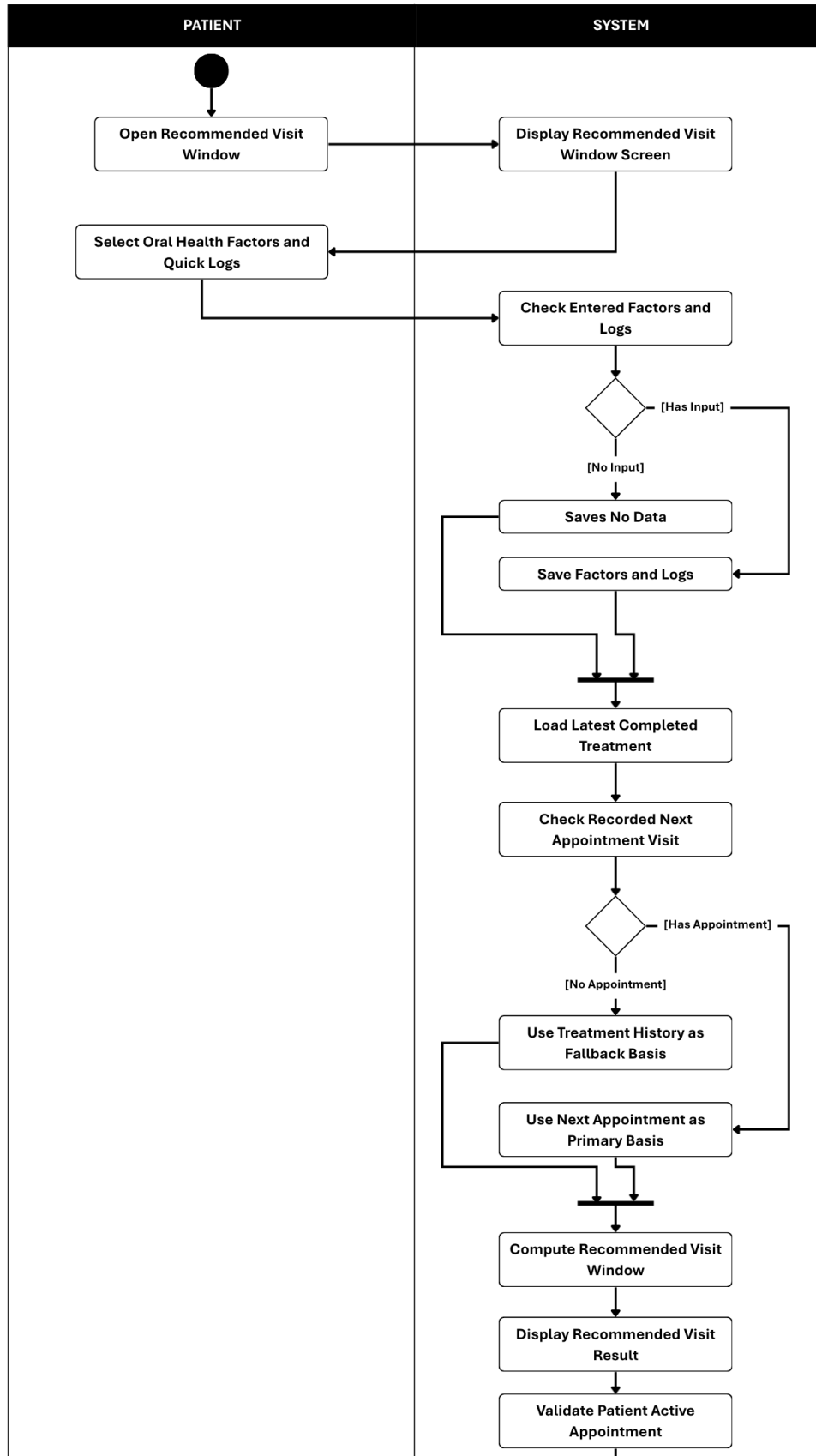


Figure 31. Oral Health Management Activity Diagram: Web

Figure 42 illustrates the web oral health management process. The system displays oral health information and random oral health facts, and the patient may review the content, book the next visit, or navigate to another section.



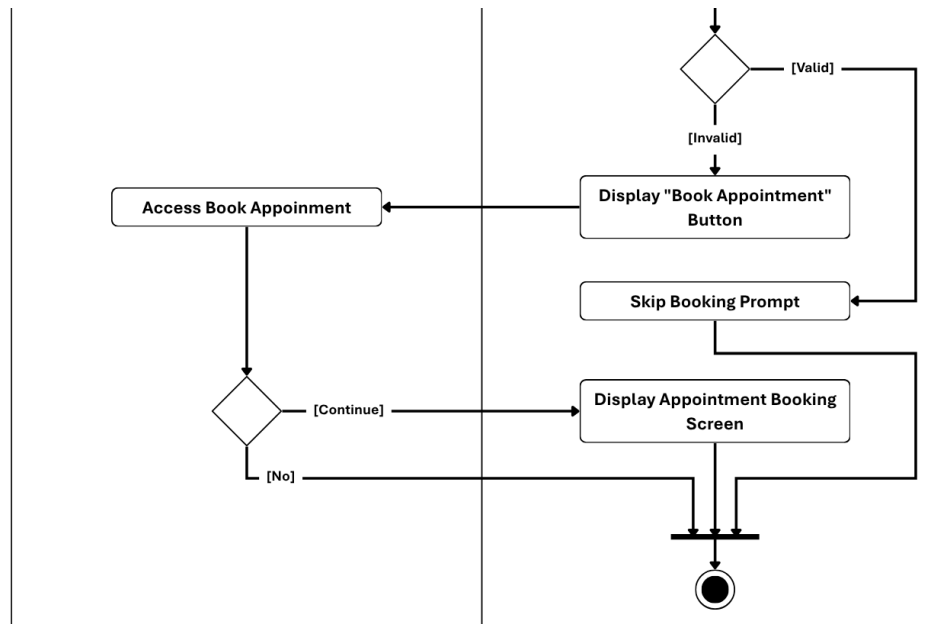
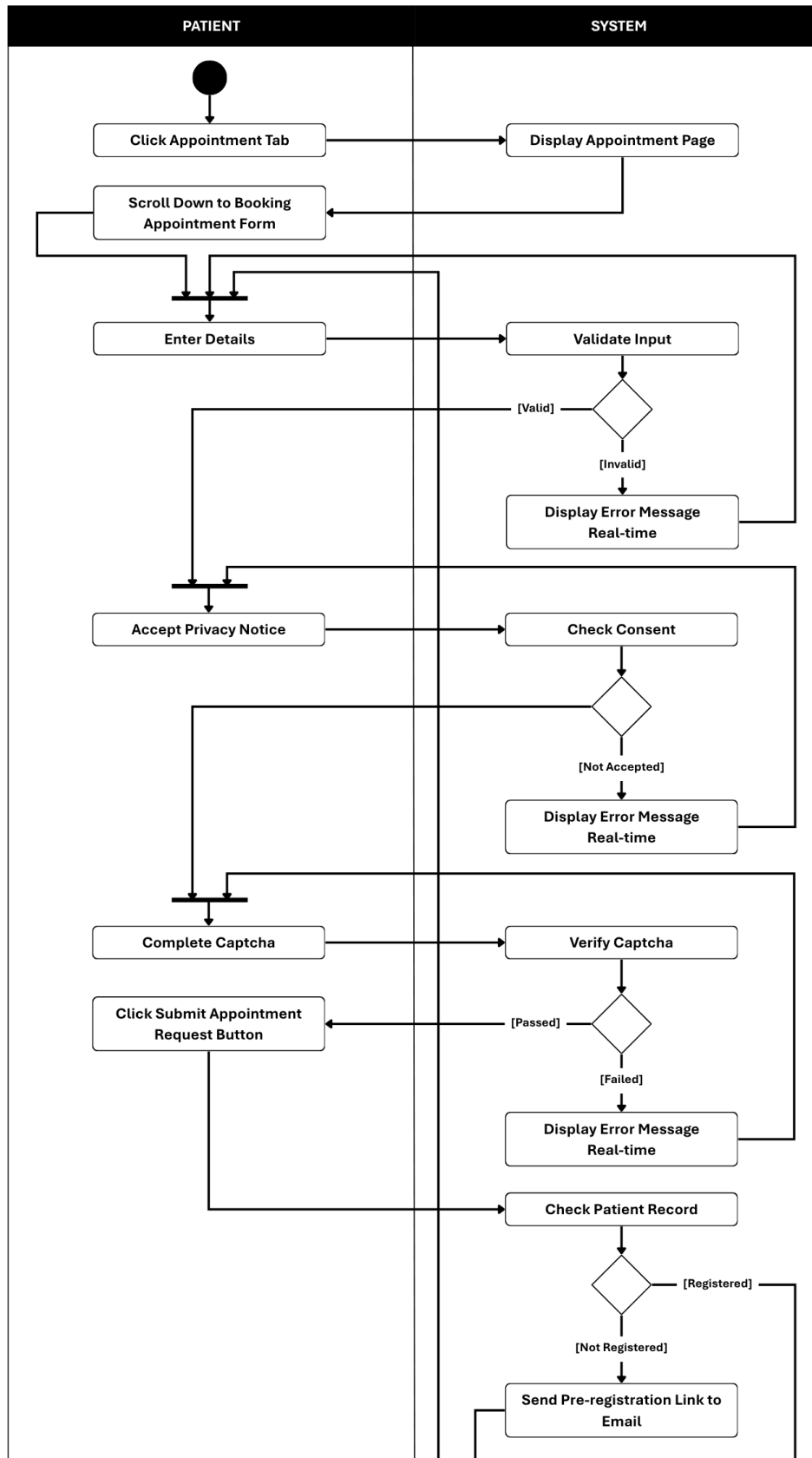


Figure 32. Recommended Visit Window Activity Diagram: Web

Figure 43 illustrates how the web recommended visit feature computes a recommended visit window. The patient enters oral health factors and quick logs, and the system checks saved logs, completed treatments, appointment history, and recorded next-appointment details before displaying the recommended visit result and an optional booking prompt.



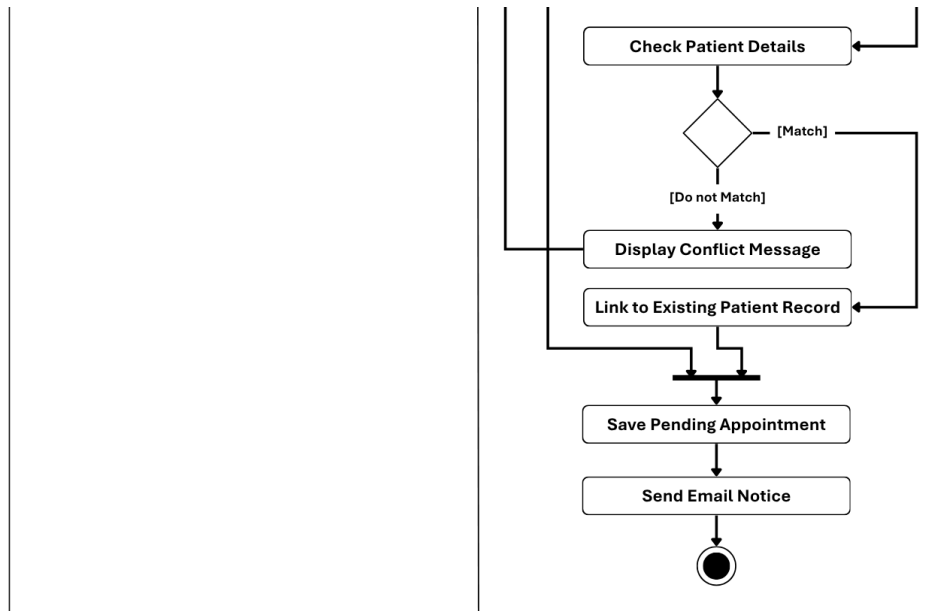


Figure 33. Website Booking Activity Diagram: Web

Figure 44 illustrates the public website booking process. The patient opens the appointment page, enters booking details, accepts the privacy notice, completes CAPTCHA verification, and submits the request. The system validates inputs and consent, checks patient records, handles registered or matching patient details, saves the pending appointment, sends the pre-registration link when needed, and emails the appointment notice.

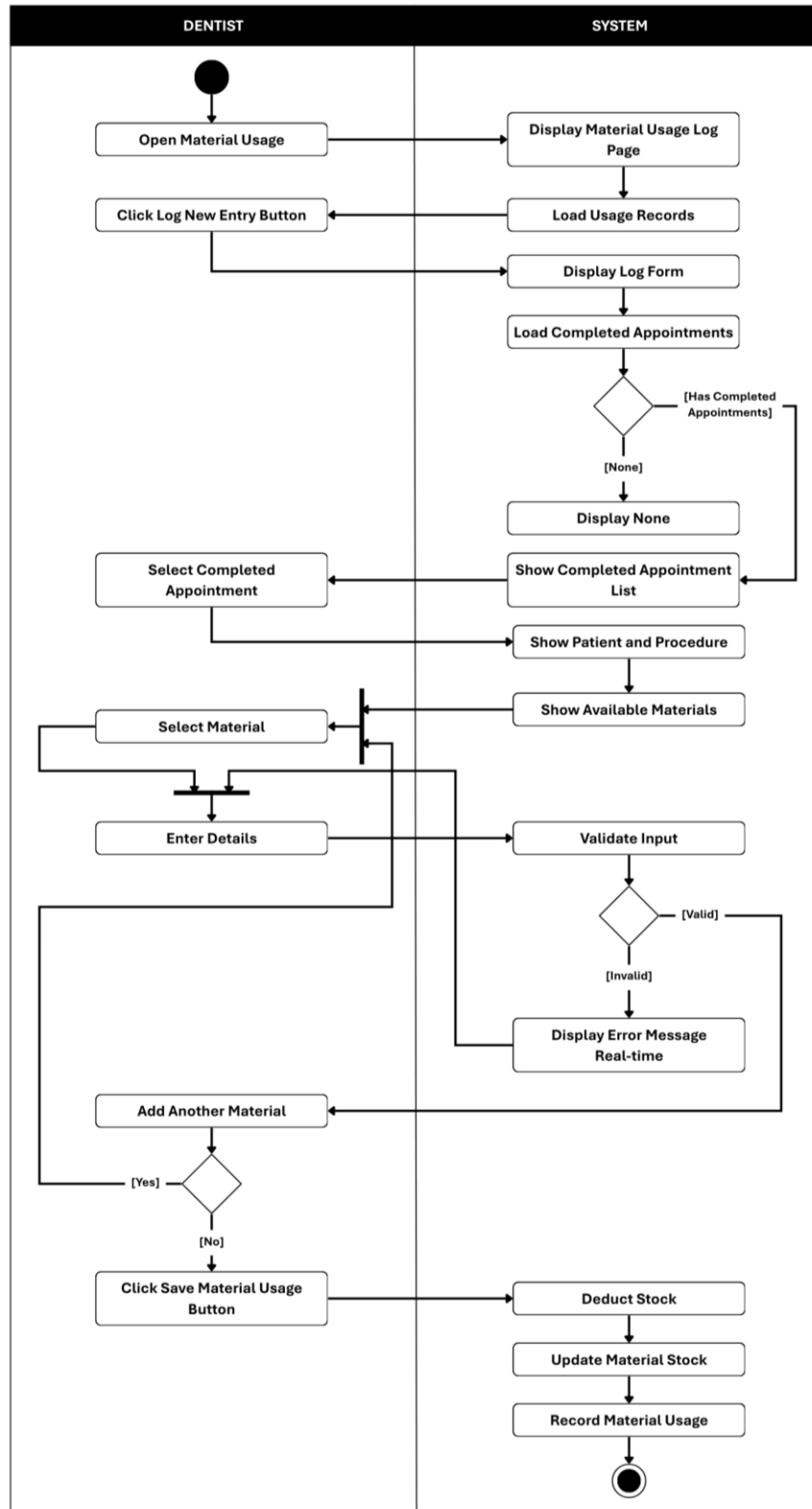


Figure 34. Material Usage Logging Activity Diagram: Web

Figure 45 illustrates how dentists log material usage during completed appointments. The dentist opens the material usage log, selects a completed appointment and materials, enters usage details, and submits the entry. The system validates the input, shows available materials, deducts stock, updates material inventory, and records the material usage.

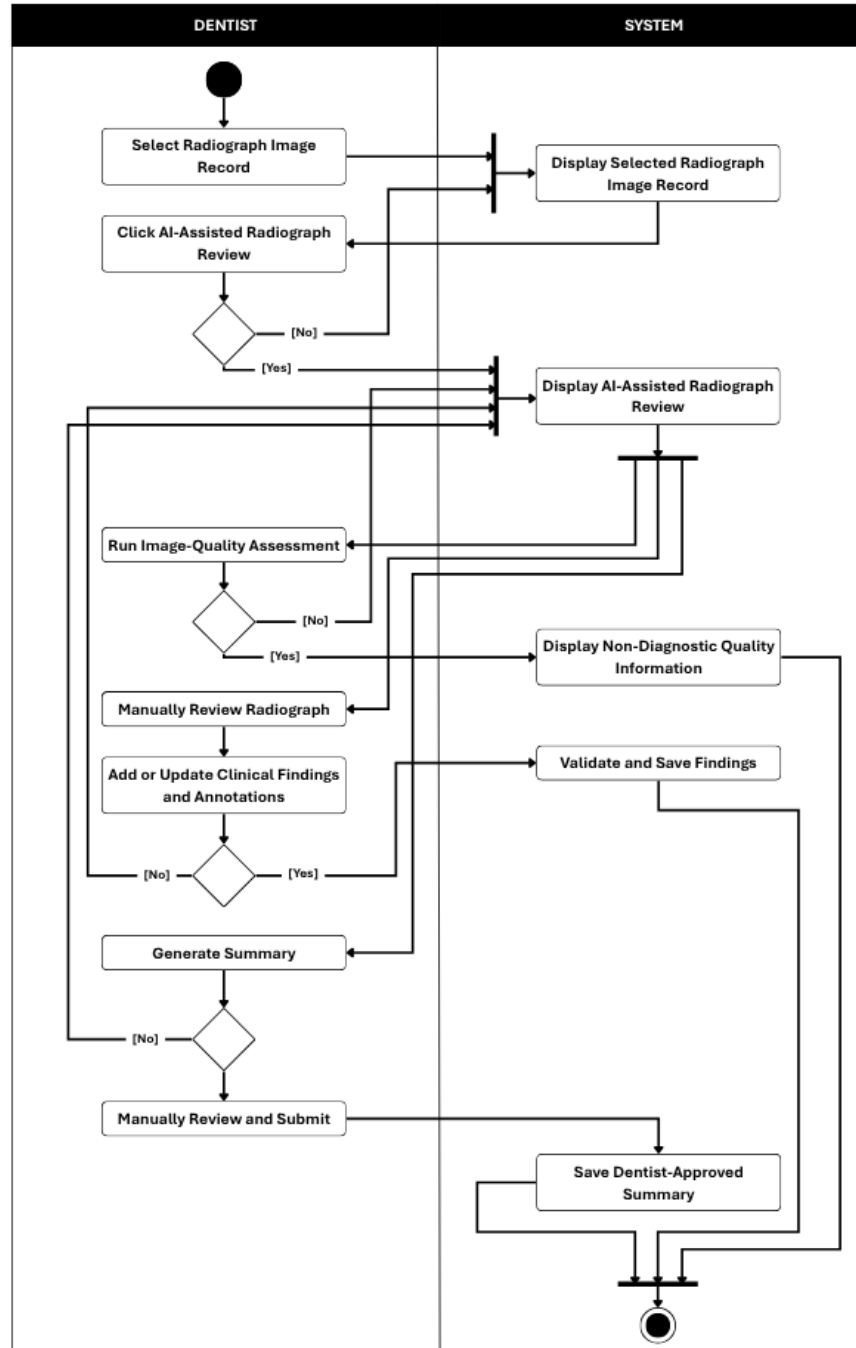
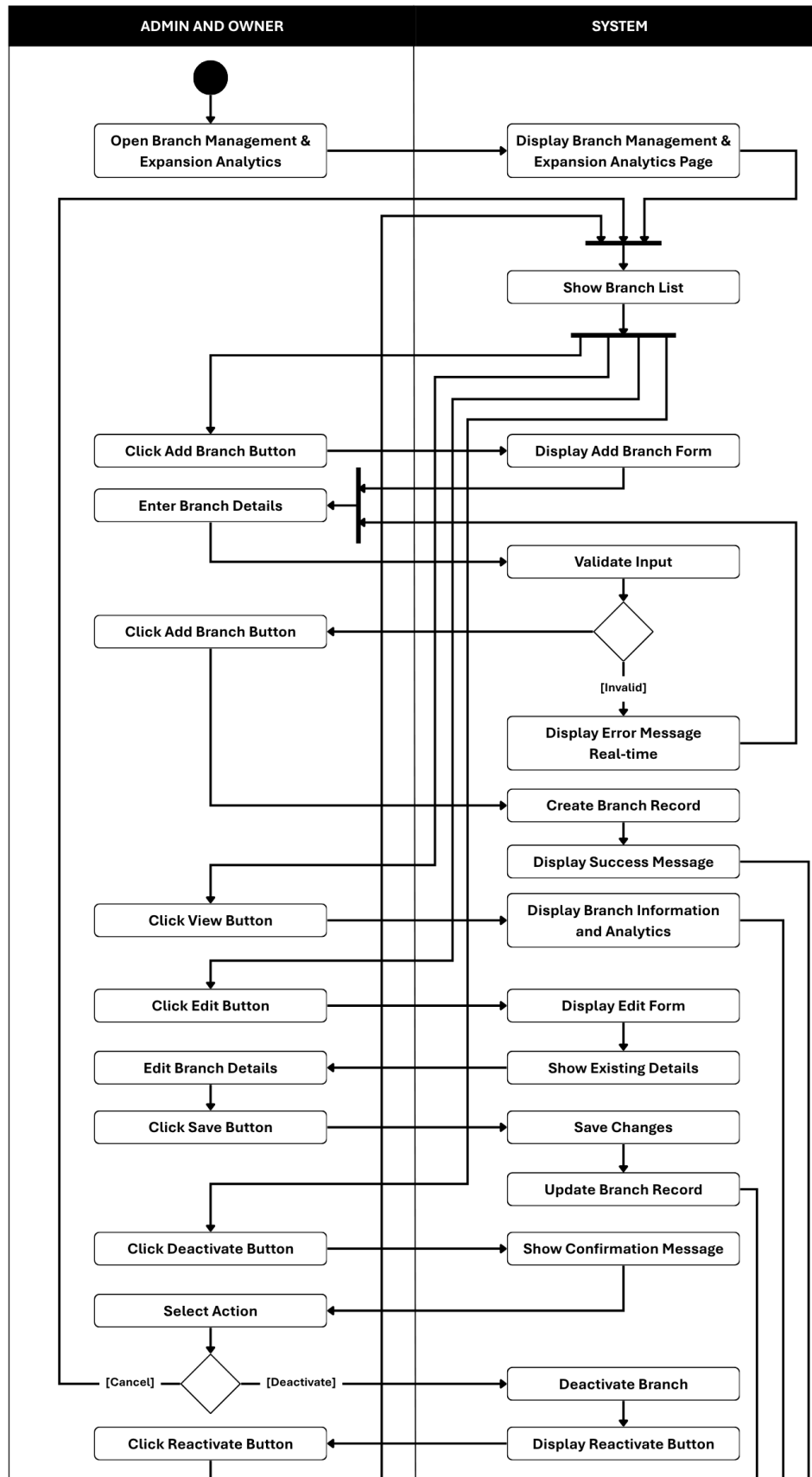


Figure 35. AI-Assisted Radiograph Review Activity Diagram: Web

Figure 46 illustrates the AI-assisted radiograph review process. The dentist selects an existing radiograph record, reviews the original image and its objective quality information, and records the relevant clinical findings. After completing the review, the dentist generates and edits a summary draft based on the verified findings and linked EMR information. The system validates the review requirements and publishes the summary to the patient's record only after the dentist

explicitly approves it. The system then notifies the patient that a dentist-approved radiograph summary is available.



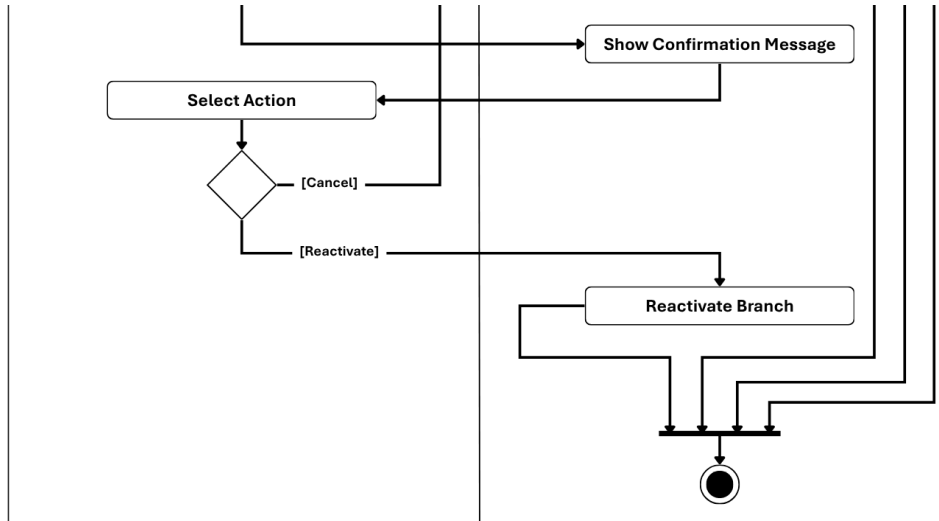


Figure 36. Branch Management and Expansion Analytics Activity Diagram: Web

Figure 47 illustrates branch management and expansion analytics. The admin or owner opens the branch module, views the branch list and analytics, adds a branch with validated details, views and edits existing branch information, saves updates, and deactivates or reactivates branches through confirmation steps.

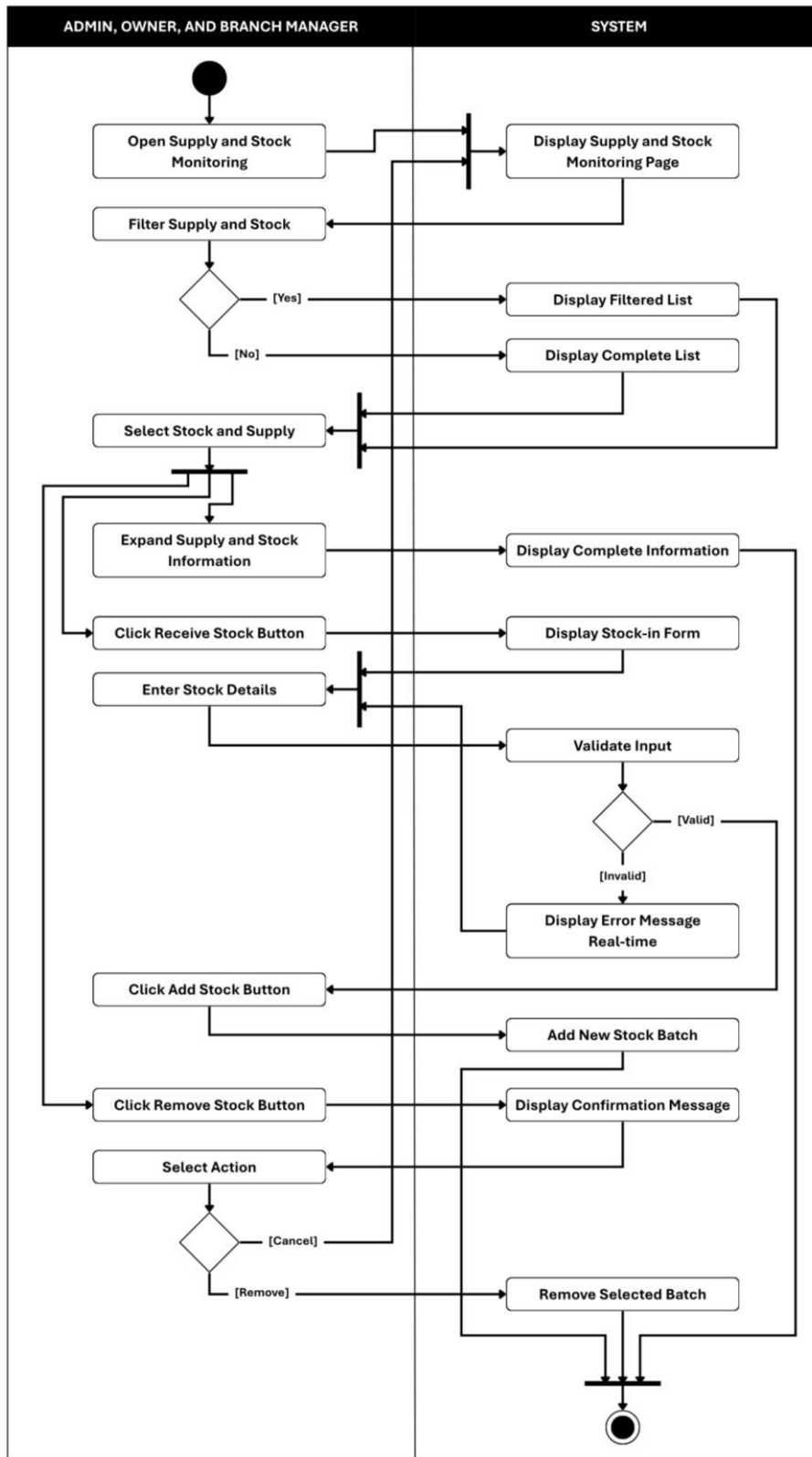


Figure 37. Supply and Stock Monitoring Activity Diagram: Web

Figure 48 illustrates the supply and stock monitoring process. Authorized users open the supply and stock page, filter and view supply records, expand supply information, receive new stock batches, and remove selected batches; the system validates entries, updates stock records, and displays confirmation messages.

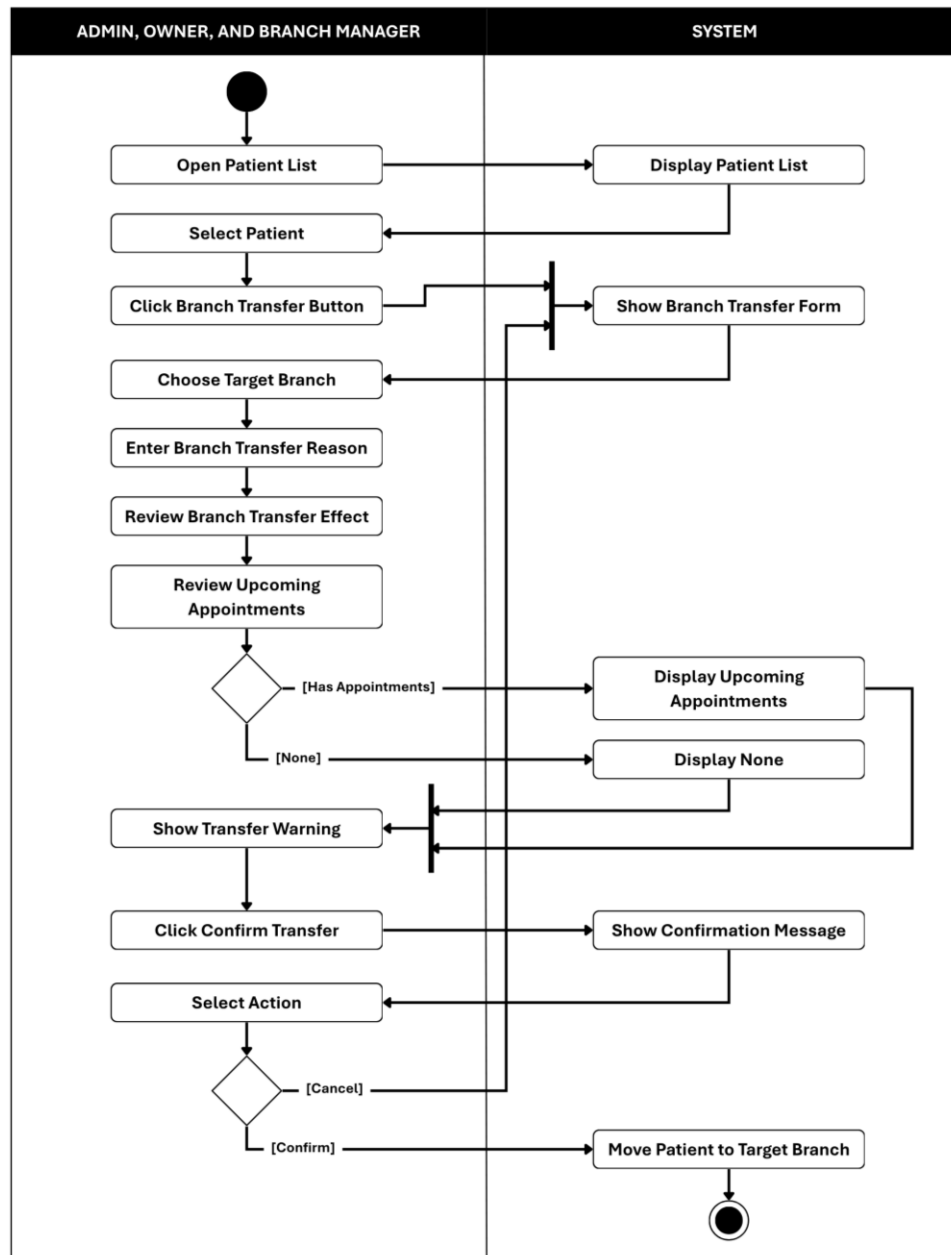


Figure 38. Branch Transfer Management Activity Diagram: Web

Figure 49 illustrates how a patient is transferred to another branch. The authorized user selects a patient, opens the branch transfer form, chooses the target branch, enters a transfer reason, reviews upcoming appointment effects and warnings, confirms the action, and the system moves the patient to the target branch.

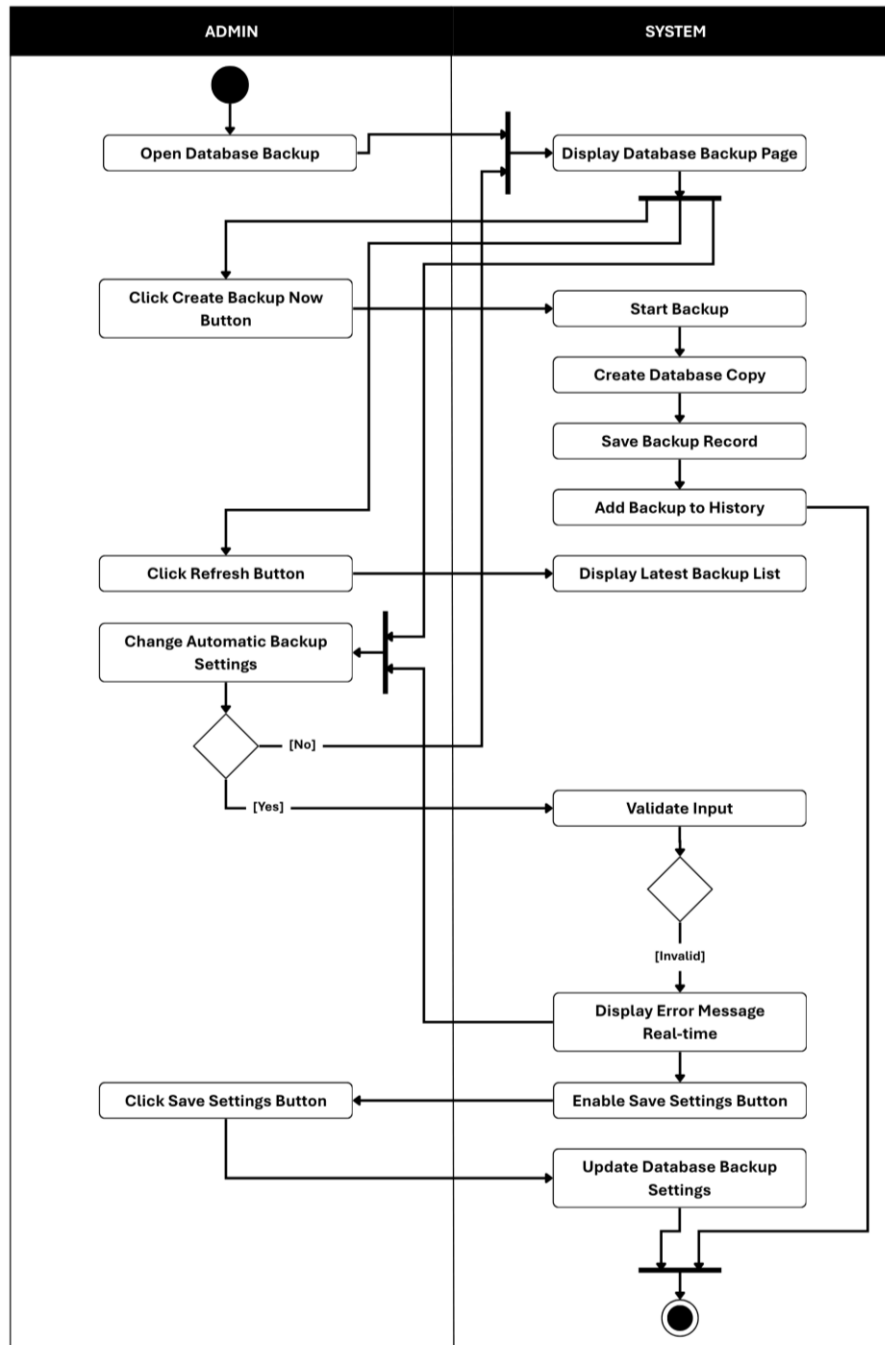


Figure 39. Database Backup Activity Diagram: Web

Figure 50 illustrates the database backup process. The admin opens the backup page, creates a manual backup, refreshes the backup list, or changes automatic backup settings. At the same time, the system starts the backup, creates a database copy, saves the backup record, updates the backup history, validates settings, and saves configuration changes.

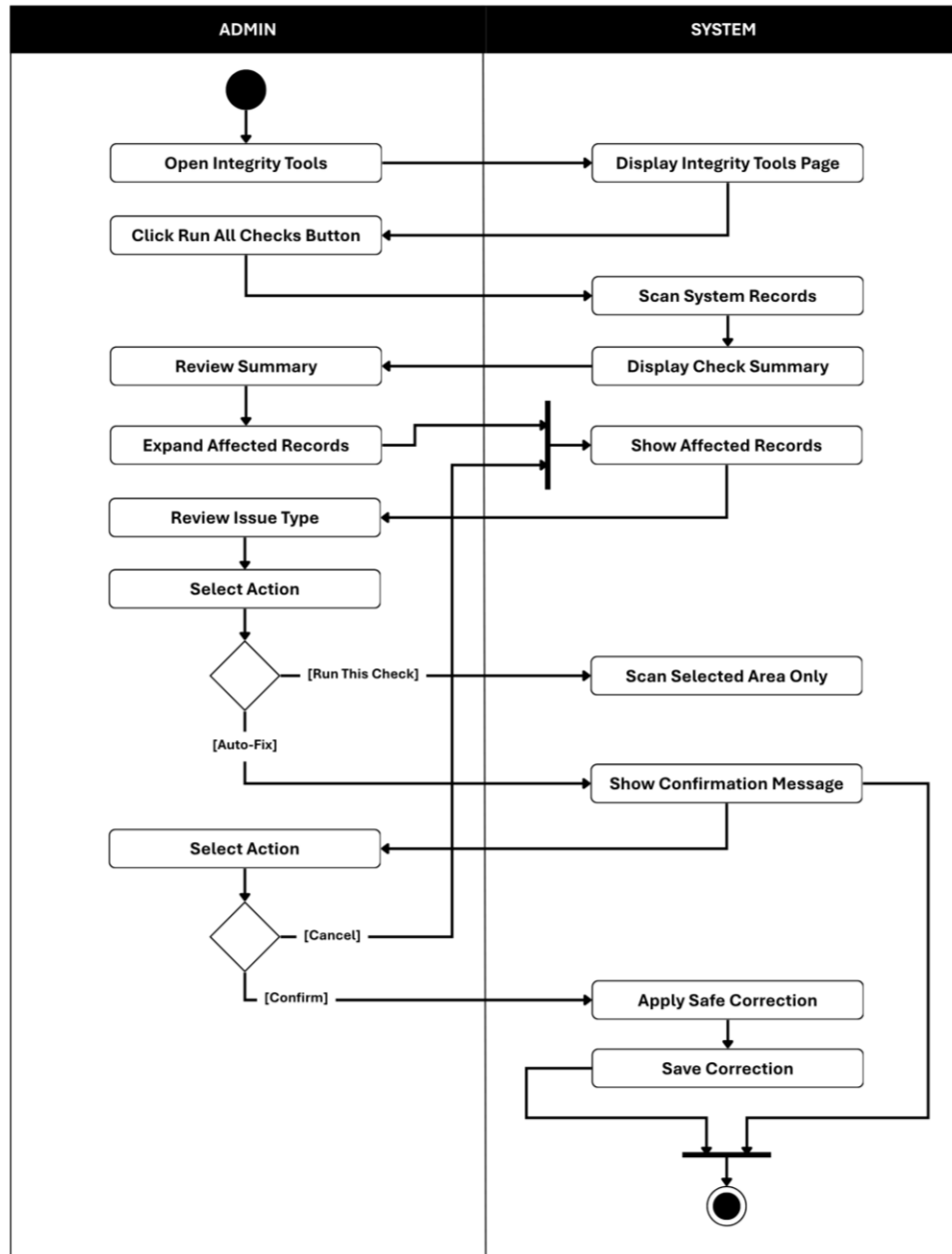
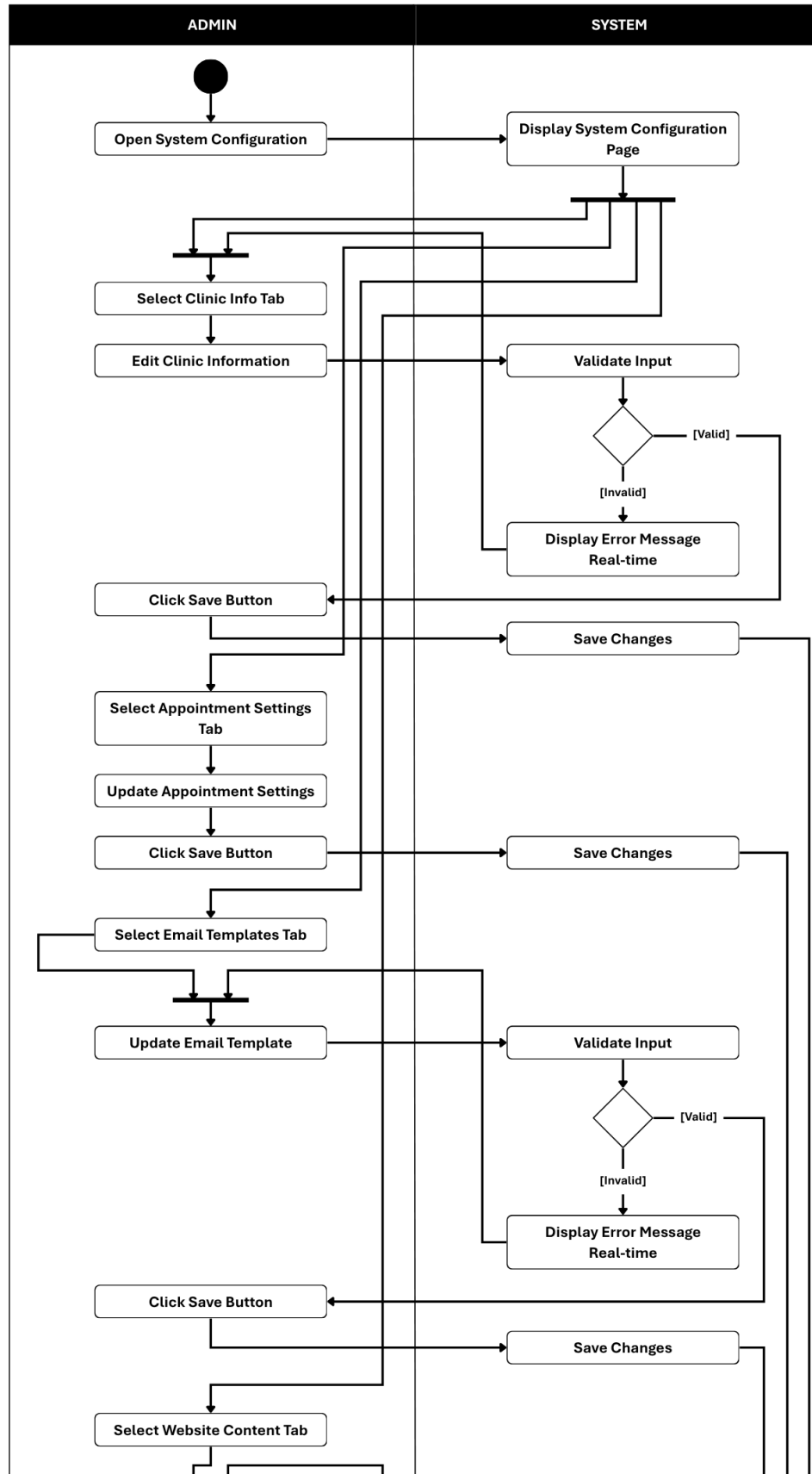


Figure 40. Integrity Tools Activity Diagram: Web

Figure 51 illustrates how the admin uses integrity tools. The admin runs all checks or scans a selected area, reviews the summary and affected records, selects a correction action, confirms the change, and the system applies and saves the safe correction.



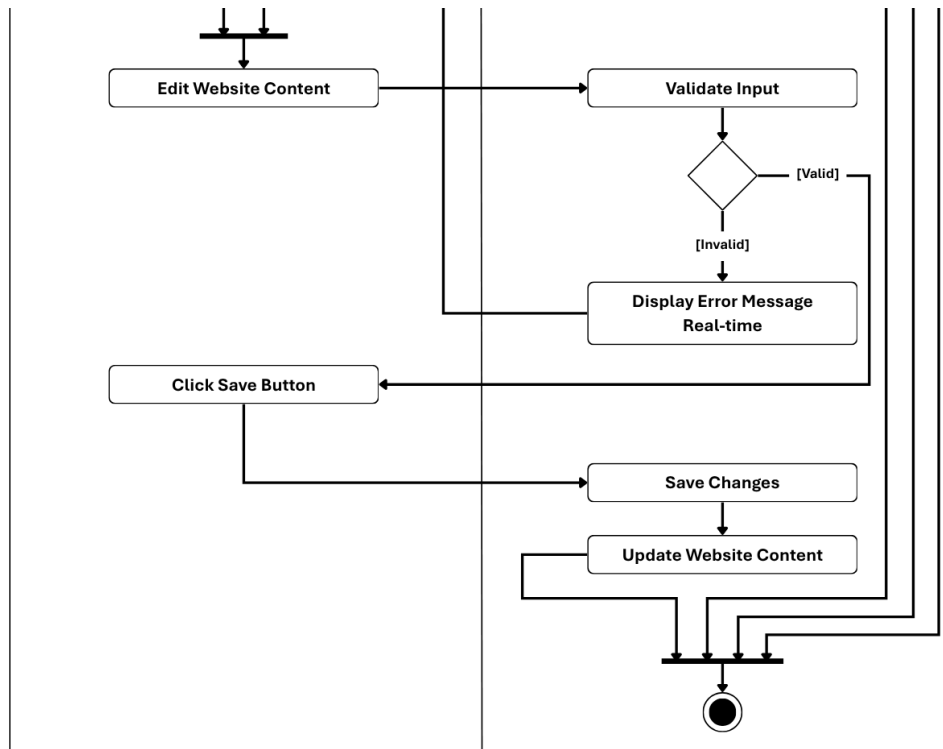


Figure 41. System Configuration Activity Diagram: Web

Figure 52 illustrates the system configuration process. The admin opens System Configuration and may update clinic information, appointment settings, email templates, or website content. The system validates each input, displays real-time errors for invalid entries, saves valid changes, and updates the corresponding configuration records.

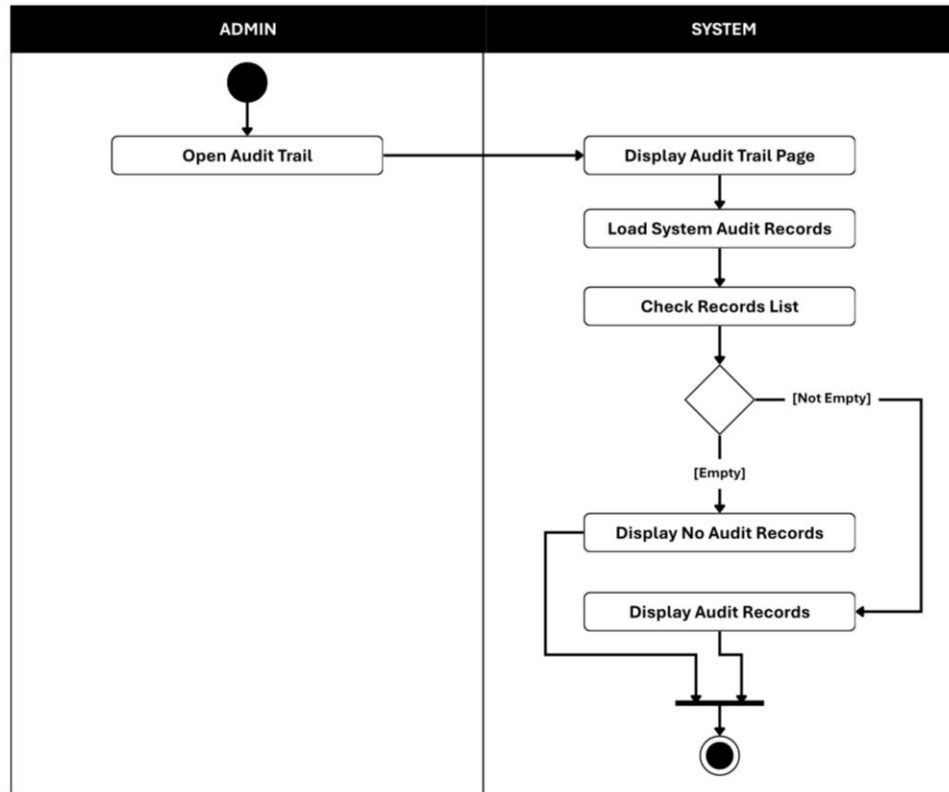


Figure 42. Audit Trail Activity Diagram: Web

Figure 53 illustrates the audit trail process for the admin. The admin opens the Audit Trail module, and the system displays the audit trail page, loads system audit records, and checks whether records are available. If records exist, the system displays the audit records; otherwise, it shows that no audit records are available.

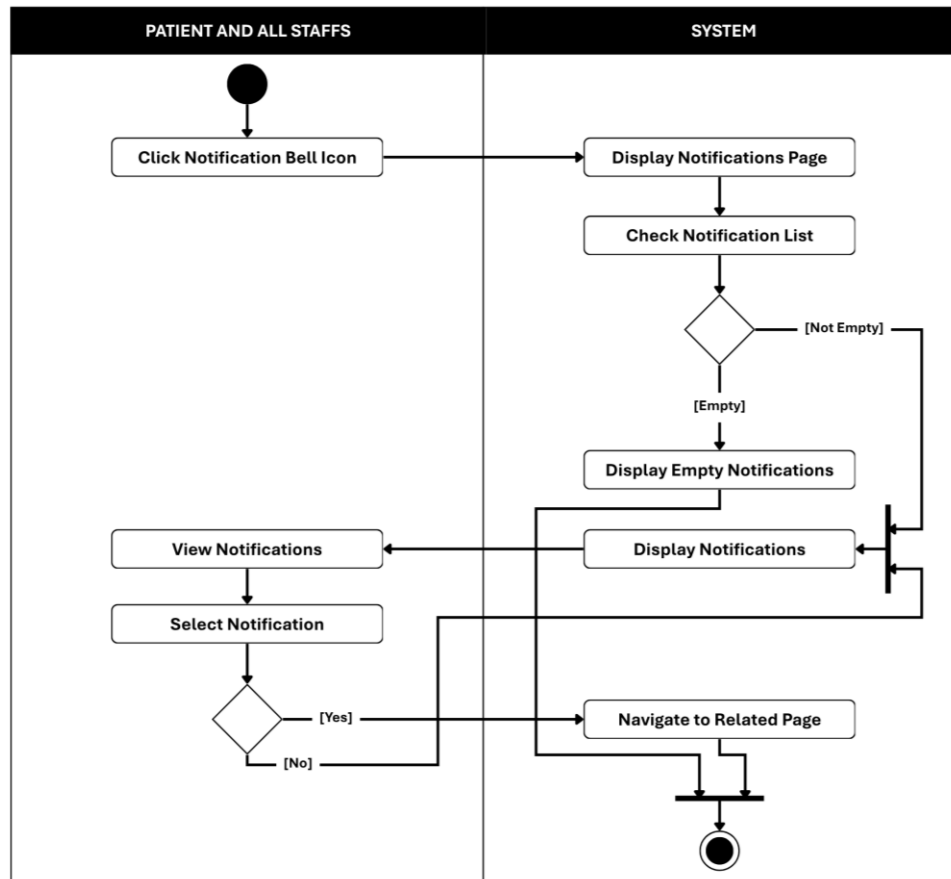


Figure 43. Notifications Activity Diagram: Web

Figure 54 illustrates the notification process for patients and staff users. The user clicks the notification bell icon, and the system displays the notifications page, checks the notification list, and shows either an empty notification message or available notifications. When the user selects a notification, the system may redirect the user to the related page.

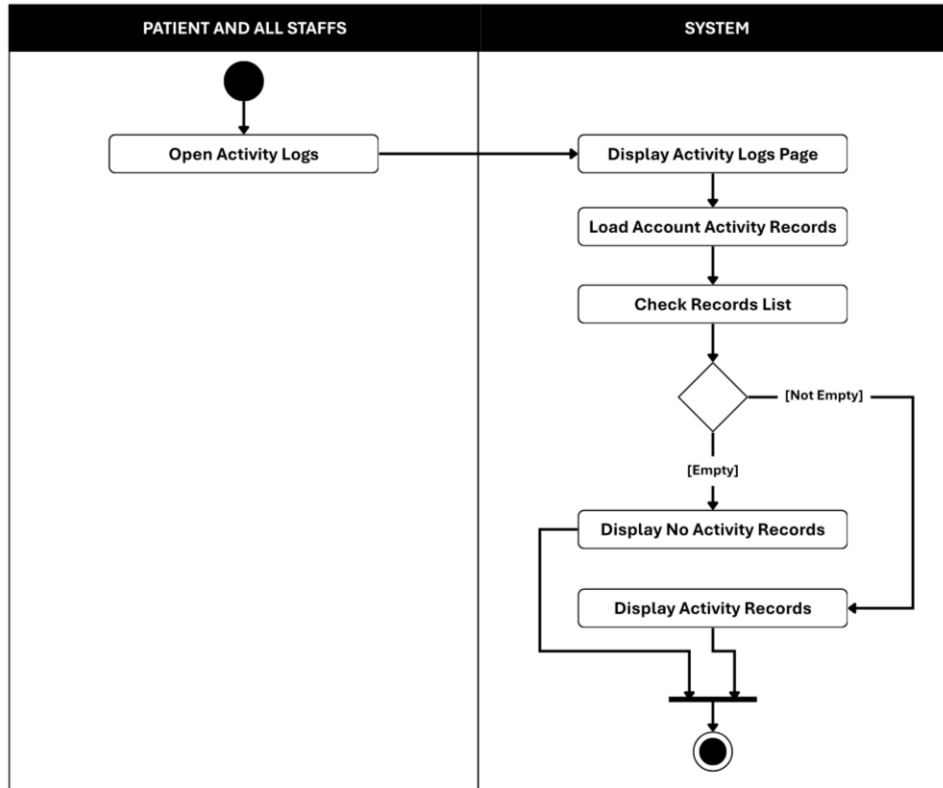
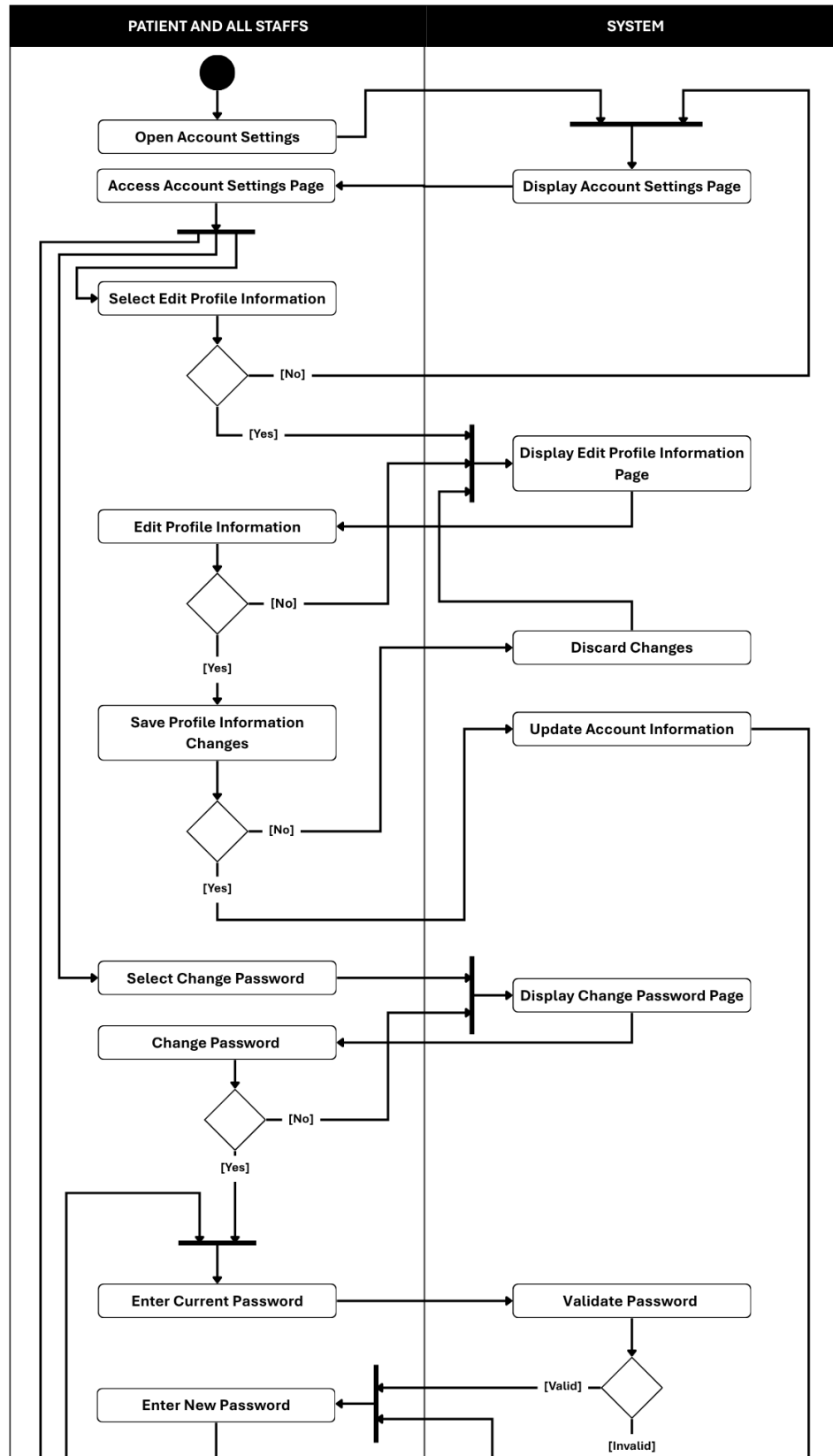


Figure 44. Activity Logs Activity Diagram: Web

Figure 55 illustrates the activity logs process for patients and staff users. The user opens the Activity Logs page, and the system loads account activity records and checks whether activity records are available. It displays either the activity log list or a message indicating that no activity records exist.



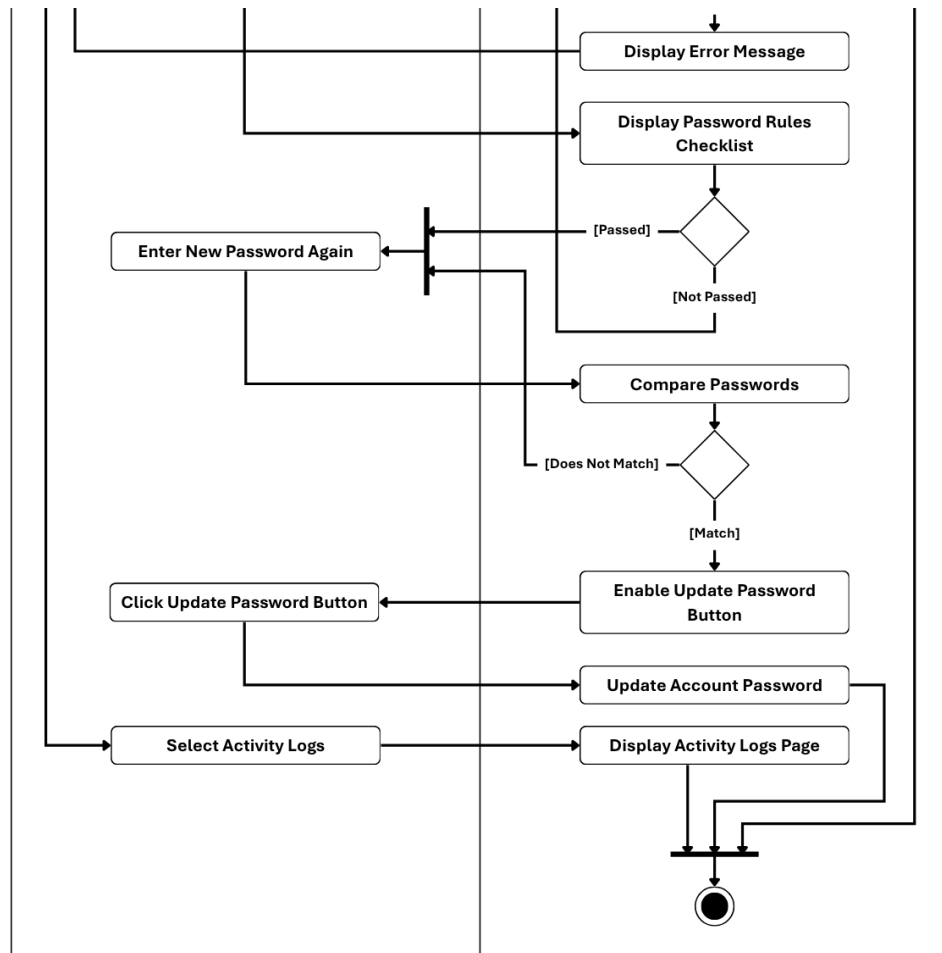


Figure 45. Account Settings Activity Diagram: Web

Figure 56 illustrates the account settings process for patients and staff users. The user opens Account Settings and may edit profile information, change the password, or access activity logs. The system displays the account settings page, validates profile or password changes, shows errors and password rules when needed, enables the update button once requirements are met, saves valid changes, and updates the account information.

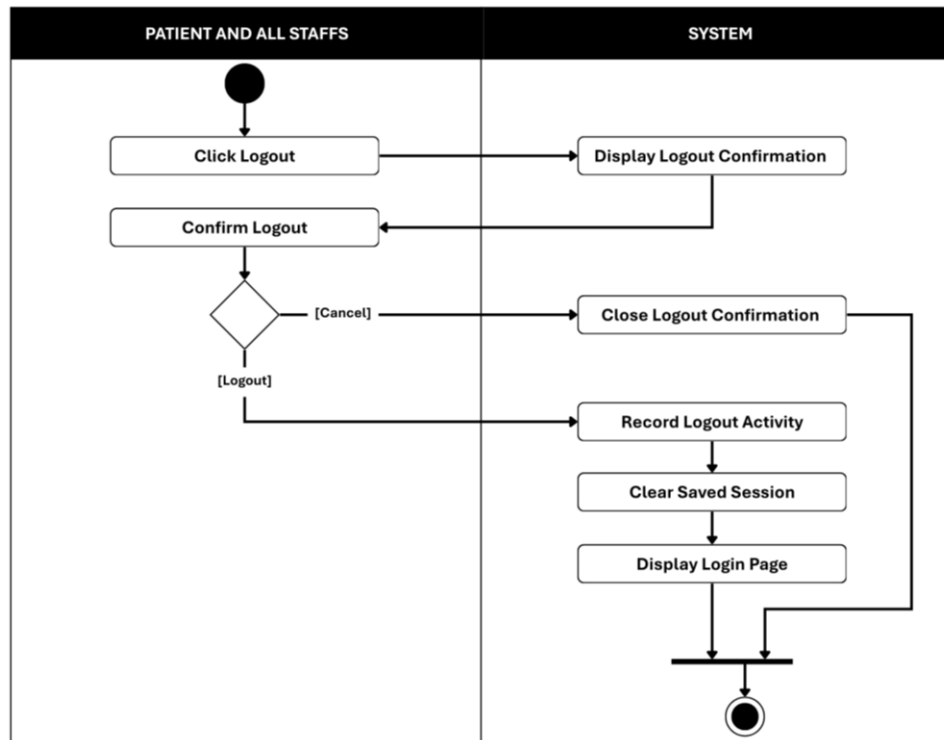


Figure 46. Logout Activity Diagram: Web

Figure 57 illustrates the logout process for patients and staff users. The user clicks Logout and confirms the action. If the user cancels, the system closes the confirmation dialog. If the user proceeds, the system records the logout activity, clears the saved session, and redirects the user back to the login page.